

Variable implicativity in *enough* constructions: causation, coercion, and composition

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Abstract

Enough constructions (*x be ADJ enough to P*) license a variable, aspect-sensitive pattern of complement inferences, often mimicking the inferential profile of complement-entailing *implicative* verbs (Karttunen, 1971; Hacquard, 2005). This paper examines English and French *enough/assez* constructions, arguing that complement inferences arise via a conspiracy of lexical, compositional, and contextual features. Building on von Stechow et al. (2004), I offer a semantic analysis on which *enough* composes with a one-place predicate *P*, a gradable adjective ADJ and an entity *x* by attributing to *x* a degree of ADJ which is *necessary* for *P(x)*. Implicativity arises when the relationship between the matrix property and the *enough* complement is understood causally: i.e., where context supports the assumption that having a particular degree of ADJ is causally determinative for *P(x)*. In certain cases, this assumption has a grammatical dimension: I argue that when ADJ introduces a *dispositional* property with a causal relationship to *P(x)*, complement entailments arise automatically in grammatical contexts which demand episodic interpretations. I propose that these contexts trigger a special type of **aspectual coercion** (Moens and Steedman, 1988; de Swart, 1998) which applies selectively to dispositional statives and returns a predicate of *evidentiary* actions understood as sufficient cause for the *enough* complement. The analysis makes good on Hacquard’s (2005) suggestion that *enough*’s inferences are genuine instances of implicativity, aligning *enough* with Nadathur’s (2023b) causal account of lexical implicatives; looking ahead, it suggests a connection between implicativity, causal meaning, and the aspect-sensitive *actuality entailments* of ability modals (Bhatt, 1999).

1 Introduction

As first described by Karttunen (1971), verbs in the **implicative class** are characterized by a pattern of complement entailments, demonstrated below for *manage* and *dare*:

- (1) a. Mallory **managed** to summit Everest. → *Mallory summited Everest.*
b. Mallory did not **manage** to summit Everest. → *Mallory did not summit Everest.*
- (2) a. We **dared** to enter Mammoth Cave. → *We entered Mammoth Cave.*
b. We did not **dare** to enter Mammoth Cave. → *We did not enter Mammoth Cave.*

The inferences in (1)-(2) suggest a close relationship—logical equivalence—between an implicative assertion and its reconstructed complement, but it is intuitively clear that the two claims are not interchangeable. The contrast between them is due to *not at-issue* (projective) content introduced

by the implicative: *manage* is often taken to presuppose that its complement requires effort from the sentential subject (Karttunen and Peters, 1979, a.o.), while *dare* presupposes a need for boldness.

While the semantics of implicatives has engendered significant interest (Coleman, 1975; Karttunen and Peters, 1979; van Leusen, 2012; Baglini and Francez, 2016; Nadathur, 2023b; White, 2019; Zuber, 2020, a.o.), investigations often sideline a related class of ‘ambiguously’ implicative predicates (Karttunen, 1971, pp.354–356). This class includes phrasal *enough* predicates, as in (3):

- (3) a. Nur was clever enough to solve the riddle. $\leadsto$ *Nur solved the riddle.*
 b. Nur was clever enough to join the chess club. $\not\leadsto$ *Nur joined the chess club.*¹

Like a lexical implicative, *be clever enough* conveys that a property of the sentential subject is involved in the (potential) realization of the complement. Despite this similarity, *enough* constructions do not fully replicate the inferential profile of verbs like *manage* and *dare*. (3a) seems to default to an implicative reading (as shown above), but its complement is defeasibly implicated, rather than entailed; following Karttunen, (3b) provides an assessment of Nur’s mental capacity without implying either the truth or falsity of the *enough* complement (see also Meier 2003).²

The idea that certain *enough* predicates are genuinely implicative is nevertheless strengthened by Hacquard (2005), who observes that the complement inferences of cognate French *assez* constructions become full entailments when the matrix predicate is inflected for perfective aspect. The complement of (4), for instance, cannot be consistently denied:

- (4) Nur a été assez intelligente pour résoudre l’énigme, #mais elle n’a pas
 Nur has be.PP enough intelligent for solve.INF the-riddle, #but she NEG-has NEG
 résolu l’énigme.
 solve.PP the-riddle.

‘Nur was-PFV intelligent enough to solve the riddle, #but she did not solve-PFV the riddle.’³

Together, (3a) and (4) motivate an account on which *enough/assez* constructions share at least a subset of the semantic components responsible for lexical implicatives’ inferential profiles: indeed, Hacquard uses data like (4) to argue that *enough* predicates have the complement-entailing semantics of ‘true’ implicatives. Full unification faces several analytical challenges, however: it does not immediately explain either the defeasibility of implicativity in English claims like (3a) nor the non-implicativity of (3b). Moreover, full implicativity for *enough* requires overt perfectivity. As (5) shows, the imperfective counterpart of (4) patterns with (3a) in permitting complement denial:

- (5) Nur était assez intelligente pour résoudre l’énigme, mais elle n’a pas
 Nur be.PST.IPF enough intelligent for solve.INF the-riddle, but she NEG-has NEG
 résolu l’énigme.
 solve.PP the-riddle.

‘Nur was-IPF intelligent enough to solve the riddle, but she did not solve-PFV the riddle.’

¹The symbol $\leadsto$ indicates an *invited inference* (i.e., a default but defeasible implicature). I use $\not\leadsto$ to mark the absence of a default; this should not be taken as an indication that the inference in question is blocked in all contexts.

²*Too* predicates (e.g., *be too stupid*) are also optionally implicative (Karttunen, 1971), showing the same inference patterns as *enough* predicates (*modulo* polarity reversal between matrix and complement). I do not discuss *too* constructions here for reasons of space, but see Meier (2003), Hacquard (2005), and Nadathur (2017, 2023a).

³The PFV annotation in this and subsequent translations corresponds to the French *passé composé*, which combines a present auxiliary (*avoir*) with the past participle (PP) of the main verb. *Passé composé* has largely supplanted the simple past (*passé simple*) as the perfective marker in spoken French (see, e.g., Hacquard 2009).

Finally, the inference patterns of lexical implicatives contrast with those of *enough* predicates in both English and French (Hacquard, 2006, 2020; Nadathur, 2017, 2023a). The English contrast can be seen in the comparison between (1)/(2) and (3a); (6) shows that the French implicative *réussir* (‘succeed’, ‘manage’) does not share the aspect sensitivity of *être assez intelligent(e)* (‘be intelligent enough’), instead licensing complement entailments regardless of aspectual marking.⁴ This difference is unexpected if *enough* predicates are ‘at base’ implicative.

- (6) a. Nima **a réussi** à s’enfuir, #mais il ne s’est pas enfuir.
 Nima has succeed.PP at REFL-flee.PP, #but he NEG REFL-is NEG flee.PP
 ‘Nima managed-PFV to escape, #but he did-PFV not escape.’
 b. Nima **réussissait** à s’enfuir, #mais il ne s’est pas enfuir.
 Nima succeed.PST.IPF at REFL-flee.PP, #but he NEG REFL-is NEG flee.PP
 ‘Nima managed-IPF to escape, #but he did-PFV not escape.’

The data in (1)-(6) present a complex picture. On the one hand, there is a strong intuitive connection between the interpretation of *enough* constructions and that of lexical implicative claims, and this link is supported by French data like (4). At the same time, there is some optionality in *enough* predicates’ implicativity, and comparing (4)-(5) to (6) shows that the semantics of *enough* predicates and lexical implicatives cannot be fully equivalent. A satisfactory account should handle not only the variable strength of *enough* inferences, but also their distribution: a unified analysis of lexical and *enough* implicativity must simultaneously explain the convergence of (1)/(2) and (4), the non-implicativity of (3b), defeasibility in (3a) and (5), and locate the source of aspect sensitivity (seen in 4-5) in some semantic feature of *enough* predicates which ‘true’ implicatives lack.

This paper aims to meet the above challenges. Like Hacquard (2005), I treat the entailment in (4) as genuinely implicative—i.e., as reflecting the presence of the same components that drive lexical implicatives’ entailments. I diverge from Hacquard in arguing that these components are not hard-coded in *enough*, but instead emerge in the combination of compositional semantic structure and context-driven inference. I take as my starting point the analysis of lexical implicatives proposed in Nadathur (2023b), on which their inferential profile follows from presupposed relationships of *causal necessity* and *sufficiency* between a lexically-specified prerequisite and the realization of the complement, but argue that *enough* has a necessity-only lexical semantics which varies in modal flavour (Meier, 2003; von Stechow et al., 2004). As a result, *enough* constructions license implicative inferences when the matrix adjective, its (modal) relationship to the *enough* complement, and aspectually-relevant features of the context conspire to replicate the background causal structure of a lexical implicative, in particular by supporting the contextual causal sufficiency of some action (linked to the matrix adjective) for complement realization.

The paper is structured as follows. §2 provides an overview of Nadathur’s (2023b) causal approach to implicative semantics. §3 examines empirical variation in *enough* claims, arguing that defeasible implicativity depends on the possibility of interpreting the *enough* assertion as describing a sufficient cause for the complement proposition. §4 addresses the temporal and aspectual dimensions of *enough* implicativity, focusing in on a special set of cases in which the at-base stative assertions of *enough* constructions with *dispositional* (actionable) matrix adjectives (e.g., *fast*, but not *tall*) are coerced (Moens and Steedman, 1988; de Swart, 1998)—under perfective marking—into

⁴ As a reviewer notes, this claim oversimplifies: (6b) has a coherent interpretation just in case the *imparfait* (IPF) receives a progressive (rather than habitual) interpretation. This does not obviate the aspectual contrast between lexical implicatives and *assez* predicates: the latter lack complement entailments under both progressive and non-progressive readings, while lexical implicatives lack them only in the former case (see §3.6).

manifestations of the properties they attribute: where the context already supports a causal role for the matrix property, this reinterpretation fully neutralizes the contrast between *enough* predicates and lexical implicatives, producing the convergent pattern first noted by Hacquard (2005). §5 summarizes the main proposals, and concludes by offering directions for future investigation.

2 Implicative inferences as causal entailments

To pursue the idea that *enough* complement inferences represent instances of *implicativity*, we must first establish a semantic characterization of this property. In this section, I use Nadathur’s (2023b) account of implicativity to identify three semantic features which lead to the target inferential profile, and discuss the consequences of this approach for the analysis of *enough*.

2.1 The architecture of implicative meaning

In addition to the characteristic entailments shown in (1)-(2), an account of lexical implicativity must explain the discursive non-equivalence of an implicative assertion and its complement, attributed by Karttunen (1971) to projective (presuppositional or *not at-issue*) content.

- (1) a. Mallory managed to summit Everest. → *Mallory summited Everest.*
b. Mallory did not manage to summit Everest. → *Mallory did not summit Everest.*
- (2) a. We dared to enter Mammoth Cave. → *We entered Mammoth Cave.*
b. We did not dare to enter Mammoth Cave. → *We did not enter Mammoth Cave.*

While there is disagreement as to the precise content of *manage*’s presuppositions (see, e.g., Coleman 1975; Karttunen and Peters 1979; Baglini and Francez 2016), Nadathur (2023b) suggests that the picture is clearer with verbs like *dare* or *condescend*, shown below: in each case, the implicative specifies a particular requirement or *prerequisite* for complement realization.

- (7) a. We dared / did not dare to enter Mammoth Cave.
→ *Daring (courageous) action was required to enter the cave.*
- b. The president condescended / did not condescend to meet with the petitioners.
→ *Suppressing disdain or disinterest was required to meet with the petitioners.*

The need for the prerequisite (its status *qua* prerequisite) and its characterization are evidently projective. It is the polarity of an implicative claim, however, that determines whether the prerequisite was satisfied: thus, as (8) demonstrates, prerequisite satisfaction is at least part of the *at-issue* content of an implicative statement.

- (8) a. We dared to enter Mammoth Cave (#but we didn’t have the courage).
→ We were daring/behaved courageously (and entered the cave).
- b. We did not dare to enter Mammoth Cave (#but we had the courage).
→ We were not daring/did not behave courageously (and did not enter the cave).

The pattern in (8) suggests that implicative prerequisites are understood to be both necessary and sufficient for complement realization. This pattern of presupposition and assertion matches the semantic schema first proposed by Karttunen (1971, p.352):

Let v stand for any implicative verb and S for the [...] complement [...]. [If] $v(S)$ constitutes the central part of the proposition [...] the semantic analysis of the whole sentence can be represented by the following schema:

- (9) *Presupposition:* $v(S)$ is a necessary and sufficient condition for S
Proposition: $v(S)$

Karttunen straightforwardly predicts the inference patterns of verbs like *dare* as long as the “central part of the proposition”—i.e., the implicative assertion—is taken to be the realization of the relevant (lexically-specified) prerequisite. A positive implicative claim asserts $v(S)$, satisfying a sufficient precondition for S , and generating positive complement entailments when presupposition and assertion are taken together. Conversely, a negative claim asserts the falsity of $v(S)$, indicating that a necessary precondition for S fails, and thereby deriving the failure of S .⁵

Within this schema, Nadathur proposes that *manage* is underspecified or semantically bleached: it is implicative in that it indicates the relevance of *some* prerequisite for complement realization, but differs from *dare* and *condescend* in failing to characterize this prerequisite. Non-specificity can be recruited to explain the apparent malleability of *manage*’s projective content, which has been argued to vary at least between inferences of effort, intention, and unlikeliness. (10) readily licenses all three inferences, but Baglini and Francez (2016; 11a-11b) and Coleman (1975; 11c) show that any of these inferences can be felicitously blocked, as long as some other projective material remains intact. (11a) plausibly presupposes intention, but not effort or unlikeliness; (11b) presupposes effort and intention without unlikeliness, and (11c) presupposes unlikeliness, but not effort or intention.

- (10) Mallory managed/did not manage to summit Everest.
a. *Licenses:* Summiting Everest was effortful for Mallory.
b. *Licenses:* Mallory intended to summit Everest.
c. *Licenses:* It was unlikely that Mallory would summit Everest.
- (11) a. Clad in civilian clothes [...], they easily managed to get back over the Volga.
b. Now it’s becoming obvious that Fork will manage to kill someone important.
c. Harry spent all evening trying very hard not to insult Ursula, but he managed to insult her, all the same.

The malleability of *manage*’s projective content is predicted if it presupposes something relatively abstract which can be resolved more precisely on the basis of context. Thus, if the core presupposition is simply that some condition (or set thereof) is necessary and sufficient for complement realization, context can support an interpretation on which the positive resolution of this condition requires effort, is intentionally linked to the complement, or on which successful resolution is *a priori* unlikely. We might then expect a speaker to select implicative *manage* when the discourse context supports the existence of a potential obstacle to complement realization (cf. Karttunen, 2014), but where no available verb provides an appropriate characterization of the action needed to overcome this obstacle (or where the speaker is unable or unwilling to commit to such a characterization).

⁵See Nadathur (2023b) for additional English and Finnish data which support the generalizability of this scheme.

2.2 Causal dependence in implicative semantics

2.2.1 Motivating a role for causal dependence

Karttunen’s schema provides the right framework for implicative semantics, but it can be precisified. In particular, (9) does not indicate whether the key relationships of necessity and sufficiency should be interpreted alethically—making an implicative claim and its complement logically if not discursively equivalent—or with respect to a particular modal flavour. Building on Baglini and Francez (2016), Nadathur (2023b) argues that lexical implicatives invoke *causal* relationships, requiring that the prerequisite acts as a *causally necessary* and *causally sufficient* condition for complement realization. The consequence of this restriction is that implicatives are a type of causal predicate: we expect them to be used just in case (a) an event in which the implicative prerequisite is satisfied concretely precipitates a causal chain which brings the complement about, and (b) prerequisite failure blocks all causal pathways from the reference time situation to complement realization.

While I will not fully justify this restriction here—relevant arguments are offered in the references cited above—(12) provides support for the causal approach by showing that the salience of a *deontically* necessary and sufficient condition does not (by itself) license the use of *manage*.

- (12) a. *Context.* In the United States, being 18 years old is a legally necessary and legally sufficient condition for male citizens to register for selective service (you are prohibited from registering before you are 18, and obligated to register thereafter). Juniors in U.S. high schools are typically between 16 and 18 years old; suppose that the information that Khalid is a high school junior is shared in the utterance context, but that while the speaker is aware of his precise age, the addressee may not be.
- b. Khalid managed to register for selective service (??by virtue of turning 18).

The claim in (12b) need not be infelicitous, but it cannot make assertoric reference to the deontic prerequisite: a speaker who knows that Khalid has just turned 18 cannot use (12b) to assert this information. Assuming that prerequisite satisfaction is part of *manage*’s at-issue contribution, the intended use of (12b) should be possible if Khalid’s age constitutes an acceptable prerequisite: however, (12b) instead suggests that Khalid overcame a concrete obstacle to registration: for instance, that he successfully navigated a complex registration process. In other words, (12b) is acceptable only insofar as it suggests that Khalid resolved conditions which stand *causally prior* to registration, supporting the claim that *manage* invokes a causal background.

Combining the causal restriction with schema (9) produces Nadathur’s implicative template:

- (13) **Causal semantics for implicatives** (Nadathur, 2023b, p.343). Given an implicative verb *I*, an individual argument *x*, and a one-place predicate *P*, the proposition *I*(*x*, *P*):
- a. *Presupposes:* the existence of an eventive one-place predicate *A* such that *A*(*x*) (the realization of *A* by *x*) is *causally necessary* and *causally sufficient* for the realization of *P*(*x*) in the utterance context.
- b. *Asserts:* the realization (occurrence) of *A*(*x*)

Implicative verbs differ in what they lexicalize about the nature of a prerequisite *A*(*x*), but uniformly presuppose its existence and causal relationship to complement *P*(*x*). Proposal 13 derives the characteristic inferential profile of lexical implicatives, including complement entailments, projective content, and polar entailments about the status of the prerequisite (see, e.g., 8). The prerequisite presupposition (13a) blocks discursive equivalence between implicative claims and their reconstructed complements; prerequisite satisfaction is asserted (see 13b), handling (polarity-sensitive)

prerequisite entailments. Finally, the status of the complement follows as a (causally) entailed consequence of combining presupposition and assertion. The specification of causal flavour for the key necessity/sufficiency relations explains data like (12), restricting the use of lexical implicatives to contexts in which the licensing prerequisites are causally-determinative precursors to the complement, and thus have the effect of concretely precipitating (or precluding) complement realization.

2.2.2 Causal modality

Nadathur formalizes the causal relations in Proposal 13 in terms of Schulz’s (2011) framework for *causal entailment* in a deterministic **structural equation model** (SEM; Pearl, 2000). The details are not essential here, so I provide only a sketch of the main ideas, and refer the interested reader to Nadathur (2023b, pp.322–330) for a more complete exposition and an illustration of how causal necessity and sufficiency (defined via SEMs) derive the desired interpretations for implicative verbs.

For current purposes, what matters is that a causal SEM is a formal structure in which valid patterns of causal inference can be characterized: a model encodes causal dependencies between a set of propositions corresponding intuitively to the occurrence or non-occurrence of certain events.⁶ More precisely, an SEM D contains a directed acyclic graph $G = \langle V, E \rangle$ (where $E \subset V \times V$) over a set V of unvalued propositional variables; the existence of a directed edge leading from variable X to variable Y ($\langle X, Y \rangle \in E$) indicates that (a valuation of) X causally influences (the valuation of) Y .⁷ A *dependent* variable Y —i.e., any variable which occurs as the second coordinate of some element of E —can be the endpoint of multiple directed edges in G , so that SEMs allow us to represent caused eventualities as the consequences of collections of causes acting together. The model D pairs graph G with a set F of *structural equations* expressing causal laws: for each dependent variable Y , F specifies precisely how distinct valuations of Y ’s immediate ancestors determine Y ’s valuation.⁸

Given a situation s (a set of facts, corresponding to a partial true-false valuation of the set V), the causal laws derive a set of consequences or *causal entailments* (Schulz, 2011). An SEM thus sets up an inferential system which parallels classical entailment, but in which the logical concept of “what follows” from certain premises is translated into a notion of “what results” from a situation in the causally normal course of events. Causal necessity and sufficiency are defined using causal entailment: assuming that Y is downstream of X ($X, Y \in V$), a given valuation $\langle X, v_1 \rangle$ is sufficient for a valuation $\langle Y, v_2 \rangle$ ($v_1, v_2 \in \{0, 1\}$) in a situation s just in case the situation obtained by adding $\langle X, v_1 \rangle$ to s causally entails $\langle Y, v_2 \rangle$ (i.e., guarantees $\langle Y, v_2 \rangle$ as a causal consequence), and $\langle X, v_1 \rangle$ is necessary for $\langle Y, v_2 \rangle$ in s just in case all causally consistent extensions of s that verify $\langle Y, v_2 \rangle$ also verify $\langle X, v_1 \rangle$ (less formally: the only consistent causal pathways from s to $\langle Y, v_2 \rangle$ travel ‘through’ validation of $\langle X, v_1 \rangle$; see also Nadathur and Lauer 2020; Nadathur 2023b).

⁶The underlying assumption is that causal reasoning is a language-independent cognitive capacity. As a component of world knowledge, causal information can act as an input to linguistic description (as, for instance, in the evaluation of counterfactual claims; see, e.g., Schulz 2011; Kaufmann 2013; Ciardelli et al. 2018): recent work on the use of causal models for lexical semantics typically treats these models as a discourse parameter, subject to reference and manipulation by the usual linguistic means (see Nadathur and Lauer, 2020; Baglini and Bar-Asher Siegal, 2020; Alonso-Ovalle and Hsieh, 2021; Nadathur and Filip, 2021, a.o.). Hobbs (2005), Wolff (2007), Thomason (2014), and Copley and Harley (2015) discuss alternatives to SEMs; see also Copley and Wolff (2014) for a comparative overview.

⁷If $G = \langle V, E \rangle$ is part of a deterministic SEM, a valuation of $Y \in V$ is an assignment of 0 or 1 to Y . If we let $y = \langle Y, 1 \rangle$, y corresponds to the standard notion of a proposition; $\neg y$ will correspond to $\langle Y, 0 \rangle$.

⁸For instance, if Y depends on $X, Z \in V$, one of the relevant structural equations will specify a valuation for Y for each of the four possible 0-1 assignments to the pair $\{X, Z\}$. Each line of the resulting truth table can be expressed as an implicational relationship between a conjunction of two propositions (either x or its negation conjoined with z or its negation) and the proposition y or its negation.

Kaufmann (2013) outlines a procedure by which the information encoded by a structural equation model can be mapped into the premise semantics (Kratzer, 1981), allowing us to treat causal dependence as a type of modality. A situation s is treated as a set of facts corresponding to a circumstantial modal base, and the worlds that develop in closes adherence to the causal laws (i.e., which realize the most causal consequences of s) constitute the optimal set; the causal laws of a model thus implement a *stereotypical* ordering source. In the *causal premise semantics*, causal notions of necessity and sufficiency can be defined as in (14), making reference to the set $CS_D(s, w)$ of causally-optimal worlds relative to a situation (set of facts) $s \subseteq w$ and causal model D .⁹

- (14) Let W be the set of worlds, and let $D = \langle G, F \rangle$ be a contextually relevant causal model with $G = \langle V, E \rangle$ and $X, Y \in V$ such that X is ‘upstream’ of Y in G . Let s be a consistent set of propositions, and let x, y be valuations of X and Y , respectively, such that the starting situation $s \subseteq w^*$ does not causally entail x , $\neg x$, or y .¹⁰ Then:
- a. x is **causally sufficient** for y with respect to s iff $\forall w \in CS_D(s, w)[x(w) \rightarrow y(w)]$
 - b. x is **causally necessary** for y with respect to s iff $\forall w \in CS_D(s, w)[\neg x(w) \rightarrow \neg y(w)]$

I conclude this section with a preliminary lexical entry for implicative *manage* which formalizes Proposal 13 in terms of causal modality:

$$(15) \quad \llbracket \text{manage} \rrbracket^{s, w} := \lambda P_{set} \lambda x_e. (\iota A_{set}. \forall w' \in CS_D(s, w)[A(w')(x) \leftrightarrow P(w')(x)])(w)(x)$$

Since the prerequisite $A(x)$ is presupposed to be both necessary and sufficient for the implicative complement, it is necessarily the unique determinative cause of $P(x)$ in the utterance context.¹¹ This motivates the use of the ι operator (Beaver and Krahmer, 2001): (15) restricts *manage* to contexts which supply a unique one-place predicate whose realization is necessary and sufficient for the realization of the complement $P(x)$ in all causally optimal worlds. In such a context, a positive *manage* claim establishes that $A(x)$ is actually realized, and this (causally) entails the actualization of $P(x)$ as long as the evaluation world is included in the causally optimal set (see fn.9).

2.3 The semantic parameters of implicativity

Nadathur (2023b) identifies three crucial parameters for the implicative inferential profile:

- (i) the (presupposed) existence of a necessary and sufficient condition for complement realization,
- (ii) a causal restriction on the modal flavour of necessity and sufficiency,
- (iii) an assertion which resolves the truth-value status of the prerequisite (and thus ‘activates’ the necessary and sufficient condition to precipitate or preclude complement realization)

⁹ Following Nadathur (2019, 2023a), I make the additional assumption that the evaluation world for a causal modal claim is by default maximally causally normal; this means that the set of worlds picked out by $CS_D(s, w)$ necessarily contains w itself. This captures the idea that causal modality is a subtype of circumstantial modality, as well as the intuition that we treat actual events as developing in accordance with the laws of causation.

¹⁰ Definition (14) requires checking the relationship between (valuations of) X and Y in causally consistent extensions of situation s which also validate x or $\neg x$. Such worlds need not simply augment s with a value for X , but must also validate a causally normal process for arriving at a valuation of X from s : this means that causal sufficiency and necessity can hold even if simply augmenting s with a value for X is not enough to guarantee or preclude the desired value for Y . Generating an appropriate circumstantial modal base s in the premise semantics may require letting go of certain facts which preclude an entailment from s to x or $\neg x$: see Kaufmann (2013) for a revision procedure.

¹¹ If condition A is necessary for B in context c and A is also sufficient for B in c , A must be the only necessary condition for B in c ; if A is sufficient and necessary for B , it must likewise be the only sufficient condition for B in c .

Asserted content. It is harder to see how variation in parameter (iii) might be achieved, since this constitutes the entire *at-issue* contribution of an implicative. The picture becomes clearer if we compare a lexical implicative with a *have the X* construction:

- Both (17a)-(17b) establish courageousness, but they are not equivalent: (17b), unlike (17a), has a reading on which it describes only a *capacity* for courageous action, rather than the action itself. The result is that—even with causal dependency between courageous action and $P(x)$ —(17b) need not entail its complement. The difference between action and capacity is thus a means by which complement inferences might become defeasible without changes in the presuppositional structure described by (i)-(ii). To the extent that this distinction affects the *aspectual class* of a potentially-implicative predicate, variations in parameter (iii) offer an avenue for explaining the interaction between *enough* implicativity and grammatical aspect first noted by Hacquard (2005).

The implicative behaviour of *enough* constructions involves three interrelated puzzles. The **op-tionality** puzzle, due to Karttunen (1971), was illustrated by the contrast between defeasibly implicative (3a) and non-implicative (3b), repeated below from §1. Since the full implicative profile (exemplified by *manage* and *dare* in 1-2) involves a two-way inference pattern, a complete picture of *enough* implicativity requires considering what happens under negation. As (18) shows, the contrast in (3) persists under negation: continuing (18a) by affirming its complement is highly marked (if not outright contradictory), but the anti-implicative continuation in (18b) is felicitous.

- The second puzzle, due to Hacquard (2005), involves variation in the **strength** of implicative inferences. This was illustrated in the contrast between the complement-entailing French *assez* construction in (4) and its imperfective counterpart in (5), repeated below as (19a)-(19b):

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‘Nur was-PFV intelligent enough to solve the riddle, #but she did not solve-PFV the riddle.’

- b. Nur était assez intelligente pour résoudre l’énigme, mais elle n’a pas
Nur be.PST.IPF enough intelligent for solve.INF the-riddle, but she NEG-has NEG
résolu l’énigme.
solve.PP the-riddle.

‘Nur was-IPF intelligent enough to solve the riddle, but she did not solve-PFV the riddle.’

The contrast largely persists under negation. Combining negation and perfectivity in (20a) licenses an entailment to complement non-realization, but judgements of (20b) are more nuanced. Although adding negation to the imperfective claim in (5) strongly invites a negative inference (as in 18a), my consultants find complement affirmation surprising but not contradictory. This judgement aligns with Hacquard’s (2005; pp.80–81): (21)-(22) show a two-way pattern of entailment under perfective marking, and a two-way pattern of defeasible implicativity in the imperfective.

- (20) a. Nur n’a pas été assez intelligente pour résoudre l’énigme, #mais elle a
Nur NEG-has NEG be.PP enough intelligent for solve.INF the-riddle, #but she has
résolu l’énigme.
solve.PP the-riddle.

‘Nur was-PFV not intelligent enough to solve the riddle, #but she solved-PFV the riddle.’

- b. Nur n’était pas assez intelligente pour résoudre l’énigme, mais elle a
Nur NEG-be.PST.IPF NEG enough intelligent for solve.INF the-riddle, but she has
résolu l’énigme.
solve.PP the-riddle.

‘Nur was-IPF not intelligent enough to solve the riddle, but she solved-PFV the riddle.’

- (21) a. Nima a été assez rapide pour s’enfuir, #mais il ne s’est pas
Nima has be.PP enough fast for REFL-escape.INF, #but he NEG REFL-is NEG
enfui.
escape.INF.

‘Nima was-PFV fast enough to escape, #but he did not escape-PFV.’

- b. Nima était assez rapide pour s’enfuir, mais il ne s’est pas
Nima be.PST.IPF enough fast for REFL-escape.INF, but he NEG REFL-is NEG
enfui.
escape.INF.

‘Nima was-IPF fast enough to escape, but he did not escape-PFV.’

- (22) a. Nima n’a pas été assez rapide pour s’enfuir, #mais il s’est
Nima NEG-has NEG be.PP enough fast for REFL-escape.INF, #but he REFL-is
enfui.
escape.PFV.

‘Nima was-PFV not fast enough to escape, #but he escaped-PFV.’

- b. Nima n’était pas assez rapide pour s’enfuir, mais il s’est
Nima NEG-be.PST.IPF NEG enough fast for REFL-escape.INF, but he REFL-is
enfui.
escape.PFV.

‘Nima was-IPF not fast enough to escape, but he escaped-PFV.’¹³

The third puzzle is new. It involves variation in the **inferential profile** of those *enough* constructions which show some tendency towards implicativity. This is best illustrated by imperfective *assez* claims (although it is not restricted to these claims): while (23a) is compatible with readings in which its complement fails (making an implicative inference at best defeasible), (23b) licenses an entailment to the negation of its complement. The contrast between the pairs in (21b)/(22b) and (23a)/(23b), then, is evidence that *enough* implicativity can take two forms: a two-way profile in which implicativity is aspect-sensitive and a second pattern in which the negative inference, at least, is not ‘weakened’ by the imperfective. I refer to this as a ‘one-way’ pattern of implicativity.

- (23) a. Nima était assez grand pour s’enfuir, mais il ne s’est pas
 Nima be.PST.IPF enough tall for REFL-escape.PP, but he NEG REFL-is NEG
 enfui.
 escape.PP.
 ‘Nima was-IPF tall enough to escape, but he did not escape-PFV.’
- b. Nima n’était pas assez grand pour s’enfuir, #mais il s’est
 Nima NEG-be.PST.IPF NEG enough tall for REFL-escape.PP, #but he REFL-is
 enfui.
 escape.PP.
 ‘Nima was-IPF not tall enough to escape, #but he escaped-PFV.’

This section has two goals. First, I aim to link the puzzles above to parameters of variation in the compositional structure of *enough* predicates. Second, I aim to make good on a modified version of Hacquard’s implicativity hypothesis by linking key contrasts to the presence or absence of elements of Nadathur’s (2023b) implicative template. I begin by adopting a modal semantics for *enough* based on von Stechow et al. (2004), and show that this induces a necessity presupposition that renders *enough* predicates structural cousins of one-way implicative verbs. The rest of the section examines the puzzles presented above in greater detail, comparing empirical data to the predictions of the necessity semantics in order to argue for three main claims:

1. Variation in modal flavour—governed by the relationship between the matrix adjective and the complement—directly predicts the **optionality** of *enough* predicates’ implicativity (Meier, 2003). Only circumstantially-interpreted constructions can be genuinely implicative, aligning with the predictions of the implicativity hypothesis (see Section 2.3).
2. Where implicativity arises, **inferential profile** depends on the matrix adjective: specifically, on the role that the adjectival property plays in a process for complement realization. In accordance with the implicative hypothesis, we observe a two-way profile (as in 21-22) with *dispositional* matrix adjectives that characterize actions or behaviours that act as causal

¹³Hacquard’s (2005) defeasibility judgement is for the *trop* (‘too’) construction in (1), rather than the negated *enough* construction (22b). I treat this substitution as harmless, since my consultants confirm both judgements, and all available analyses treat *enough* and *too* as semantic duals (Meier, 2003; von Stechow et al., 2004; Hacquard, 2005; Schwarzschild, 2008; Nadathur, 2017, 2019; Erlewine, 2025), making (1) equivalent to (22b):

- (1) Nima était trop lent pour s’enfuir, mais il s’est enfui.
 Nima be.PST.IPF too slow for REFL-escape.INF, but he REFL-is escape.PP.
 ‘Nima was-IPF too slow to escape, but he escaped-PFV.’

preconditions for the *enough* complement. A distinct one-way profile (see, e.g., 23) emerges where *static* (non-behavioural) adjectives invoke enabling but not precipitating preconditions. There appears to be some individual variation on this point, which I note where relevant.

3. **Inference strength** variation is aspectually driven, as per Hacquard (2005). *Contra* Hacquard, however, I argue that this follows from an interaction between grammatical and lexical aspect. The baseline assertion of an *enough* construction constitutes a *stative* property attribution, subject to reinterpretation (Moens and Steedman, 1988; de Swart, 1998, a.o.) in composition with a perfective operator. In certain cases, the output of reinterpretation describes the proximate (contextually necessary and sufficient) cause of the *enough* complement, leveling the semantic contrast between stative, necessity-based *enough* predicates and even-tive, two-way lexical implicatives.

3.1 The necessity semantics

Enough constructions are consistently analyzed as modalized comparative claims (Nelson, 1980; Meier, 2003; Hacquard, 2005; Schwarzschild, 2008; Fortuin, 2013; Nadathur, 2019, 2023a; Grano, 2022; Erlewine, 2025). I adopt an approach from von Stechow et al. (2004) on which the subject’s allocation of some gradable property is compared to an amount of the same property which is needed to make the *enough* complement realizable. *Enough* itself is treated as an equative operator which embeds a universal modal taking scope over its complement, so that the paraphrase in (24) in fact expresses a semantic equivalence.¹⁴

(24) Amira is old enough to drink. $\sim$ *Amira is as old as she must be in order to drink.*

Gradable adjectives relate individuals to sets of degrees on an appropriate *scale* (Seuren, 1984; von Stechow, 1984; Kennedy, 2001). On the (simplifying) assumption that scales are isomorphic to the nonnegative real number line, positive adjectives (e.g., *old*; 25a) relate individuals to intervals of the form $[0, n]$, while their negative counterparts (e.g., *young*; 25b) relate individuals to intervals of the form (n, ∞) . Example (26) interprets a measure claim with a positive adjective.

(25) a. $\llbracket \text{old} \rrbracket := \lambda w \lambda d \lambda x. \text{age}(w)(x) \geq d$, where $\text{age} := \lambda w \lambda x. x$ ’s age in w

b. $\llbracket \text{young} \rrbracket := \lambda w \lambda d \lambda x. \text{age}(w)(x) < d \quad (\equiv \neg \llbracket \text{old} \rrbracket (w)(d)(x))$

(26) $\llbracket \text{Amira is 21 years old} \rrbracket^w = \llbracket \text{old}(21 \text{ years})(\text{Amira}) \rrbracket^w = \text{age}(w)(A) \geq 21y$

Comparative operators like *more* and *-er* express set relations between two degree predicates. (28) shows how this works for equative *as* with a modal argument, following Heim (2001). I assume that *must* in (28) is interpreted with respect to a salient modal base f and ordering source g , and abbreviate the resulting modal domain (i.e., $\text{BEST}_{g(w)}(\cap f(w))$) as $\text{OPT}_{f,g}(w)$.

(27) $\llbracket \text{as} \rrbracket := \lambda w \lambda P \lambda Q. Q(w) \supseteq P(w)$

(28) Amira is as old as she must be. $\sim$ $(\text{as } \lambda_1 (\text{MUST}_{f,g} (\text{Amira}_2 \text{ is } d_1 \text{ old}))_3 (\text{she}_2 \text{ is } d_3 \text{ old}))$

¹⁴Von Stechow et al. (2004) respond to work from Meier (2003): the two proposals derive equivalent truth conditions for the data considered in this paper (but see Grano 2022), but locate the modality of an *enough* claim differently. Von Stechow et al. build modality into the comparative *enough* operator, while Meier assumes that *enough* selects for a complement that contains a (potentially covert) modal. I adopt von Stechow et al.’s approach largely because it facilitates treating the implicative facts; Nadathur (2023a, pp.147–152) argues that this approach may fare better than Meier’s for *enough* claims with overtly modal complements (e.g., *Amira is old enough to be allowed to drink*).

- a. $\llbracket (28) \rrbracket^{w,f,g} = \{d : \text{age}(w)(A) \geq d\} \supseteq \{d : \forall w' \in \text{OPT}_{f,g}(w) [\text{age}(w')(A) \geq d]\}$
- b. *The set of degrees less than or equal to Amira's actual age is a superset of the set of degrees d such that in all optimal worlds, Amira is at least d degrees old.*

The superset in (28a) corresponds to the interval $[0, n]$, where n is Amira's actual age. The subset is calculated by considering modal alternatives in which Amira's age varies (see also Krasikova 2011): after filtering for optimality, we obtain a set containing all degrees less than or equal to Amira's age in the optimal world where she is youngest.¹⁵ As a result, (28) requires Amira's actual age to meet or exceed some minimum standard that obtains in the optimal worlds, thus expressing that she is old *enough* by this standard. On von Stechow et al.'s analysis, the complement of an *enough* claim is involved in deriving the implicit standard: the analysis in (28) is extended to the proposed equivalence in (24) by taking the *enough* complement to restrict the modal domain (akin to a goal or purpose clause in the evaluation of teleological modals). This derives the lexical entry in (29) for *enough*, producing the truth conditions in (30) for (24). This claim most naturally takes a deontic reading (invoking a circumstantial modal base and a deontic ordering source), on which it expresses that Amira's actual age meets or exceeds the legal minimum for drinking.

- (29) $\llbracket \text{enough} \rrbracket :=$
 $\lambda w \lambda P_{\text{set}} \lambda Q_{\text{sedt}} \lambda x_e. \{d : Q(w)(x)(d)\} \supseteq \{d : \forall w' \in \text{OPT}_{f,g}(w) [P(w')(x) \rightarrow Q(w')(x)(d)]\}$ ¹⁶
- (30) $\llbracket (24) \rrbracket^w = \{d : \text{age}(w)(A) \geq d\} \supseteq \{d : \forall w' \in \text{OPT}_{f,g}(w) [\text{drink}(w')(A) \rightarrow \text{age}(w')(A) \geq d]\}$
 $\sim$ *Amira's actual age is at least the age she has in the optimal drinking world where she is youngest.*

Going beyond von Stechow et al., I argue that (29) should be coupled with the presupposition that there is some threshold degree d_n that separates degrees of the matrix adjective compatible with the *enough* complement from those that are not. The existence and relevance of such a threshold is intuitive (as on the default deontic reading of 24), and can be further motivated by observing that *enough* claims are infelicitous when there is no reason to think that the matrix property should play a difference-making role with respect to complement realizability. The marked examples in (31) illustrate both sides of the effect: height is irrelevant in (31a) because no degree of height can enable lightspeed travel, and irrelevant in (31b) because no height should preclude breathing.

- (31) a. ??Nima is tall enough to travel faster than the speed of light.¹⁷
- b. ??Nima is tall enough to breathe.

¹⁵For a degree d to be in the right hand set in (28a), it must verify *Amira is d -old* in all optimal worlds. The full set is given by the intersection over $w' \in \text{OPT}_{f,g}(w)$ of the set of degrees d that are no greater than Amira's age in w (see 32): that is, the intersection of a set of intervals of the form $[0, \text{age}(w')(A)]$. We obtain a set corresponding to $[0, m]$, where m is the *infimum* (greatest lower bound) of the set of Amira's ages in the optimal worlds ($m = \inf(\{\text{age}(w')(A) : w' \in \text{OPT}_{f,g}(w)\})$; defined as long as $\text{OPT}_{f,g}(w) \neq \emptyset$). In the cases under consideration, m is the same as Amira's age in the optimal world(s) where she is youngest.

¹⁶The formula applies to both positive and negative adjectives. If Q is negative—say, *young*—the non-modal set has the form (n, ∞) , where n is the subject's actual age. The modal set intersects intervals of the same form, and (29) returns **True** just in case n is at most the subject's age in the optimal P -world where they are oldest.

¹⁷This judgement is based on *categorical* impossibility. *Enough* is felicitous where its complement is locally precluded, as long as the matrix property has the potential to make a difference with respect to complement realizability:

- (1) Shireen was tall enough to ride the rollercoaster, but she was too young to be allowed in the amusement park.

The contrast between (1) and (31a) supports the existence of a necessity condition, insofar as the former's felicity is due to the existence of a minimum height at which rollercoaster-riding is permissible.

The lexical entry in (29) does not, by itself, capture the infelicity of (31a) and (31b). To see why, consider the modal set from (28a), rewritten below as an intersection:

$$(32) \quad \bigcap_{w' \in \text{OPT}_{f,g}(w) : \text{drink}(w')(\mathbf{A})} \{d : \text{age}(w')(\mathbf{A}) \geq d\}$$

If drinking is categorically impossible (and thus impossible at all ages; cf. 31a), there are no drinking worlds in the optimal set, and (32) is empty. The truth conditions in (30) then make *Amira is old enough to drink* vacuously true, against intuition. Conversely, if drinking is possible at any age, there are optimal worlds in which Amira is arbitrarily young and still drinks, so (32) is $\{0\}$, and *Amira is old enough to drink* is again automatically true. Both judgements are precluded if *enough* presupposes the existence of a nonzero threshold degree which divides a nontrivial and complement-precluding part of the adjectival scale from a nontrivial part of the scale which is compatible with complement realization. This condition is expressed in (33):

(33) A statement of the form *x is ADJ enough to P* is felicitous at a world *w* just in case:

$$\exists d_n > 0 : d_n = \max(\{d : \forall w' \in \text{OPT}_{f,g}(w) [P(w')(x) \rightarrow \text{ADJ}(w')(x)(d)]\})$$

Since d_n exists and is unique by definition, we can rewrite the semantics of *enough* as in (34). Reference to the threshold degree d_n highlights the parallel between *enough* claims and simple ‘attributive’ measure claims like (26): a positive *enough* claim establishes that its subject meets or exceeds the degree threshold for complement realizability, while its negation leaves the subject short, thereby failing to satisfy a necessary condition for the truth of the complement.

$$(34) \quad \llbracket \text{enough} \rrbracket := \lambda w \lambda P \lambda Q \lambda x. Q(w)(x) (\iota d_n > 0 : d_n = \max(\{d : \forall w' \in \text{OPT}_{f,g}(w) [P(w')(x) \rightarrow Q(w')(x)(d)]\}))$$

Enough claims thus emerge as near relatives of one-way, necessity-only lexical implicatives, since both predicate types presuppose the necessity of some (matrix-specified) condition for their complements and assert that this condition was satisfied.

3.2 Optional implicativity and modal flavour

Unlike lexical implicatives, *enough* constructions vary in modal flavour (Nelson, 1980; Meier, 2003, a.o.). This facet of interpretation is context-sensitive: on acceptable interpretations, the matrix property plays a difference-making role with respect to the complement, satisfying presupposition (33). The deontic reading of (24) follows from the salience of age-based drinking laws, while (35) (translating 23a) has a circumstantial reading on which height determines the accessibility of an escape route (such as a high-up window).

(35) Nima was tall enough to escape.

The necessity semantics predicts *enough* implicativity to be sensitive to modal flavour. While the necessity presupposition links the matrix adjective to the possibility of the complement, this relationship holds only in the optimal modal alternatives: evaluation world *w* is thus subject to the constraints in (33) only when it is optimal. Deontic modal domains ($\text{DN}(w) = \text{OPT}_{\text{circ,deon}}(w)$) are projected from a circumstantial modal base and a deontic ordering source, and need not be realistic: we thus do not predict implicativity for constructions like (24), as shown below.

$$(36) \quad \text{a. } \llbracket \text{Amira is old enough to drink} \rrbracket^{w, \text{cr}, \text{dn}} = \text{age}(w)(\mathbf{A}) \geq d_n$$

- b. $\llbracket \text{Amira is not old enough to drink} \rrbracket^{w,cr,dn} = \text{age}(w)(A) < d_n$
 where $d_n = \max(\{d : \forall w' \in \text{DN}(w)[\text{drink}(w')(A) \rightarrow \text{age}(w')(A) \geq d]\})$

Both claims in (36) compare Amira’s actual age to the legally-specified minimum for drinking: (36a) establishes that she is permitted to drink (since her actual age is at least as great as the age she has in the deontically-optimal drinking world where she is youngest), and (36b) that she is prohibited from drinking. The evaluation world need not be deontically optimal, however—people do not always act on their legal rights, nor refrain from proscribed action—so neither example licenses an implicative entailment. This prediction is supported by the felicity of (37a)-(37b), and it is easy to find naturally-occurring deontic examples which are not implicative. For instance, (38) in its original context takes the death of its subject as given, and conveys that this occurrence violated some ideal, in view of Halpin’s age.

- (37) a. Amira was old enough to drink, but she didn’t touch a drop.
 b. Amira was not old enough to drink, but she had some wine at the party anyway.
 (38) γ Lee Halpin was [not old enough] to die for a story—even one that needed telling.¹⁸

The **optionality puzzle** thus appears amenable to an explanation based on modal flavour. Meier (2003) proposes just such a link, using it to explain ‘minimal pairs’ like (3a)-(3b). Since the modal flavour of *enough* is determined by the relationship between its complement and the matrix adjective, changing the complement can change modal flavour even when the matrix is held constant. Non-implicative (3b) most naturally takes a deontic reading, on which permission to join the chess club is (ordinarily) granted on the basis of intelligence; implicative (3a), however, is likely to be interpreted circumstantially, since cleverness is practically implicated in riddle-solving.

- (3) a. Nur was clever enough to solve the riddle. $\leadsto$ Nur solved the riddle.
 b. Nur was clever enough to join the chess club. $\nrightarrow$ Nur joined the chess club.

The contrast in (3) suggests—in keeping with the necessity semantics—that circumstantial *enough* constructions are the prime candidates for implicativity. As (39) shows, the truth conditions for a positive circumstantial claim (evaluated with respect to a circumstantial modal base and an empty ordering source, $\text{CR}(w) = \text{OPT}_{\text{circ},\emptyset}(w)$) entail the possibility but not the actualization of its complement, while a negative claim precludes complement realization: in (39a), Nima’s height meets or exceeds the escape-enabling threshold, while in (39b) his height is insufficient.

- (39) a. $\llbracket \text{Nima is tall enough to escape} \rrbracket^{w,CR} = \text{height}(w)(N) \geq d_n$
 b. $\llbracket \text{Nima is not tall enough to escape} \rrbracket^{w,CR} = \text{height}(w)(N) < d_n$
 where $d_n = \max(\{d : \forall w' \in \text{CR}(w)[\text{escape}(w')(N) \rightarrow \text{height}(w')(N) \geq d]\})$

These results predict a one-way profile for circumstantial *enough*, which matches the data: (40a) is felicitous, but (40b) is not. The circumstantial constructions in (41a)-(41b) show the same pattern:

- (40) a. Nima was tall enough to escape, but he chose not to do so.
 b. Nima was not tall enough to escape, #but he still escaped.
 (41) a. γ Although [the package was small enough to go through the letterbox], she found a note from Hermes saying it had been delivered to a safe place, namely “the window”.

¹⁸Here and elsewhere, the diacritic γ marks naturally-occurring data. I have modified (38) slightly from the original by replacing *too young* with *not old enough*. This does not change its truth conditions or affect non-implicativity.

- b. The package was not small enough to go through the letterbox, #but I put it through.

As (36) and (39) show, the necessity semantics for *enough* only derives a contrast in implicative entailments between deontic and circumstantial constructions under matrix negation: positive claims of both types are truth-conditionally compatible with non-implicative readings.¹⁹ This is a positive outcome, since the predictions in (36) and (39) align with the judgements in (37) and (40), respectively. At the same time, however, we do not yet have an explanation for the contrast in invited implicativity observed by Karttunen (1971) and Meier (2003) for minimal pairs like (3). We can, however, find a pragmatic explanation which builds on the semantics developed thus far.²⁰

On the lexical implicative template, positive inferences rely on the causal sufficiency of the matrix condition for complement realization: satisfying this condition (by performing the action associated with the matrix clause) directly triggers the complement, simultaneously guaranteeing the success of an attempt at complement realization while (in essence) constituting this attempt. For this effect to be replicated pragmatically, the utterance context for an *enough* construction must support a similar relationship between the matrix condition and the complement: i.e., it must license the inference that if the subject has at least the threshold degree of the matrix adjective, they will realize the *enough* complement. My claim is that this inference is more natural when the matrix-complement relationship is circumstantial than when it is deontic.

To see why, consider the following. The conditional relationship above can only hold in contexts where any practical (causal and/or circumstantial) non-matrix preconditions for the complement are presumed to hold, including that the subject makes an attempt at complement-realization.²¹ In such a context, satisfying the necessity condition only affects whether or not the complement is realized if we can construe it as causally relevant. At least one of two things must be true: (A) the necessity condition determines whether an attempt at complement realization will succeed (i.e., whether the attempt can in fact *cause* the complement) or (B) satisfaction of the necessity condition itself prompts (causes) the sentential subject to attempt complement realization.

If the matrix condition is circumstantially linked to the *enough* complement, we get (A) for free: no attempt to realize the complement will succeed unless the matrix condition is satisfied. Invited implicativity for circumstantial *enough* thus only requires contextual support (or perhaps the lack of a contextual challenge) for the assumption that no other necessary preconditions are

¹⁹Negated deontic *enough* claims can be used implicatively despite the lack of complement entailments. This is most natural when there is an independent reason to expect adherence to the rules: for instance, if the relevant prohibition is assumed to be subject to the on-the-spot enforcement, as with amusement park rides in (1).

- (1) ^γThe last time we visited ... [my son] was not tall enough to ride the ... Spider-Man ... ride.
Strongly suggests: My son did not ride.

²⁰Meier (2003) explains contrastive pairs like (3) by proposing that implicative *enough* constructions actually entail their complements as a result of being interpreted with respect to a *totally realistic* modal domain (containing only the evaluation world). The target inference follows, as long as the modal domain is only required to contain a complement-realizing world. Since the necessity semantics developed in §3.1 requires the modal domain to contain both complement-realizing and complement-falsifying worlds, this proposal cannot be adopted here. Meier's own semantics for *enough* (which makes *x is ADJ enough to P* semantically equivalent to *x's actual degree of ADJ is at least as great as the degree d such that, if x has d degrees of ADJ, x can P*) only require the modal domain to contain complement-realizing worlds. However, since an appeal to total realism cannot explain why deontic *enough* constructions are sometimes implicatively interpreted, an additional pragmatic route for deriving positive implicativity must still be available; the approach I take here aims to unify positive *enough* implicativity across modal flavours.

²¹Independent of modal flavour, the target conditional (if the necessity condition holds, the subject will realize the complement) of course also holds in contexts which background complement realization. I set these cases aside, since there is no implicature to be explained: it is not the use of *enough* here that invites or licenses complement inferences.

in question for the complement. If the matrix-complement relationship is deontic, however, more background work is needed: satisfying case (A) requires us to additionally assume that a deontically necessary condition (e.g., being of legal drinking age in 36a) is subject to practical enforcement which could plausibly prevent complement realization in the event that the matrix condition fails. Deontic enough constructions can readily be used in contexts where this does not apply, and these contexts will license positive implicativity only in scenario (B): i.e., if the subject's attempt at complement realization is contingent on the permissibility of the complement. This is certainly possible, but is not a plausible default assumption: we do not in general presume that individuals restrict their actions to what is officially permitted, nor that permissibility prompts action. While individual deontic enough claims can certainly be used in contexts which support and even invite implicativity, the deontic case requires more pragmatic scaffolding than the circumstantial, and is thus less likely to default to implicativity: it is simply harder to enrich a deontically necessary condition to a causal trigger for action than it is to interpret a circumstantial condition in this way.

Variation in the modal flavour of a necessity-only matrix-complement relationship thus seems to capture the optionality of implicative behaviour for English *enough*. Turning to French, however, we find that the interaction between *assez* predicates and grammatical aspect introduces complications. Under imperfective marking, deontic *assez* constructions behave as predicted by the necessity semantics: neither (42a) nor (42b) entails an implicative reading. Judgements of the corresponding perfective examples appear to be more variable. While my informants accept both (43a) and (43b), an anonymous reviewer (R) finds (43a) contradictory. The difference seems to be in how the perfectively-marked matrix clause is understood—i.e., in what different speakers take to license the choice of perfective aspect here. My consultants take (43a) to indicate that Amira became old enough to drink legally (i.e., had a birthday) during the reference time—selecting an *inchoative* reading which is compatible with the claim that Amira did not drink—and otherwise find the use of perfective aspect marked. R, on the other hand, finds that the *assez* claim in (43a) is acceptable only on an implicative reading (suggesting that the perfective somehow induces the assumption that satisfying the matrix condition triggers complement realization) and can only recover the non-implicative inchoative reading with the addition of a temporal adverbial, as in (44).²² These judgements indicate that the use of the perfective somehow requires modulating the semantics of the underlying enough construction, and—insofar as there are two distinct patterns of interpretation—suggest that there may be different routes to achieving composition. I focus my discussion on my consultants' (C) pattern of interpretation, but make note of the R pattern to the extent I am able.

- (42) a. Amira était assez grande pour boire de l'alcool, mais elle n'en
 Amira be.PST.IPF enough old for drink.INF of the-alcohol, but she NEG-of.it
 a jamais bu.
 has never drink.PP.
 'Amira was-IPF old enough to drink alcohol, but she never drank it.'
- b. Amira n'était pas assez grande pour boire de l'alcool, mais elle
 Amira NEG-be.PST.IPF NEG enough old for drink.INF of the-alcohol, but she
 buvait parfois du vin.
 drink.PST.IPF sometimes of-the wine.
 'Amira was-IPF not old enough to drink alcohol, but she sometimes drank-IPF wine.'

²²A parallel judgement is not reported for (43b), but this may simply have been an omission.

- (43) a. Amira a été assez grande pour boire de l'alcool, mais elle n'en a
 Amira has be.PP enough old for drink.INF of the-alcohol, but she NEG-of.it has
 pas bu.
 NEG drink.PP.
 'Amira was-PFV old enough to drink alcohol, but she did not drink-PFV any.'
- b. Amira n'a pas été assez grande pour boire de l'alcool, mais elle a
 Amira NEG-has NEG be.PP enough old for drink.INF of the-alcohol, but she has
 bu du vin à la fête.
 drink.PP of-the wine at the party.
 'Amira was-PFV not old enough to drink alcohol, but she drank wine at the party.'
- (44) Quand Amira a été assez grande pour boire de l'alcool, elle n'en a
 When Amira has be.PFV enough old for drink.PFV of the-alcohol, she NEG-of.it has
 pas bu.
 NEG drink.PP.
 'When Amira was-PFV (became) old enough to drink alcohol, she did not drink-PFV it.'

The circumstantial data are even more complex. Under imperfective marking, (45a)-(45b) show the one-way profile observed for the corresponding English data (40-41).²³ Perfective marking does not seem to change these judgements. One of my consultants suggests that the use of perfective (as opposed to imperfective) in (45a-i) suggests a bounded interpretation—i.e., that while Nima's height met or exceeded the escape-enabling threshold at reference time, this relationship has since changed, meaning either that Nima has shrunk (less plausible) or that the threshold for escaping has changed. This interpretation affects the possibility of escape subsequent to the reference time, but does not require (45a-i) to entail a reference time escape. (45a-ii) establishes that Nima's height was insufficient for escaping at reference time; the perfective likewise suggests that the relationship between height and escaping has subsequently changed, but this does not affect the impossibility of reference-time escape. Implicativity assessments for the perfective versions of (45b-i)-(45b-ii) are similar, although here my consultants report that the perfective more readily implies that the packet in question no longer exists than that either its size or the size of the mailslot has changed. As in the deontic data, then, we find that the perfective modulates the semantic interpretation of the underlying *assez* claim, but that this need not affect the underlying pattern of (non-)implicativity.

- (45) Aspect-insensitive one-way implicativity:
- a. i. Nima {a été / était} assez grand pour s'enfuir, mais il ne
 Nima {has be.PP / be.PST.IPF} enough tall for REFL-escape.PP, but he NEG
 s'est pas enfui.
 REFL-is NEG escape.PP.
 'Nima {was-PFV / was-IPF} tall enough to escape, but he did not escape-PFV.'
- ii. Nima {n'a pas été / n'était pas} assez grand pour
 Nima {NEG-has NEG be.PP / NEG-be.PST.IPF NEG} enough tall for
 s'enfuir, #mais il s'est enfui.
 REFL-escape.PP, #but he REFL-is escape.PP.

²³The judgements marked in (45) reflect the C pattern. R reports a non-contradictory reading for the imperfective variant of (45a-ii), but does not report the same for (45b-ii).

- ‘Nima {was-PFV / was-IPF} not tall enough to escape, #but he escaped-PFV.’
- b. i. Le paquet {a été / était} assez petit pour tenir dans la boîte
 The packet {has be.PP / be.PST.IPF} enough small for fit.INF in the box
 aux lettres (mais le facteur l’a passé par la fenêtre).
 of.the letters (but the postman it-has put.PP through the window).
 ‘The packet {was-PFV / was-IPF} small enough to fit in the mailbox (but the post-
 man put-PFV it through the window).’
- ii. Le paquet {n’a pas été / n’était pas} assez petit pour
 The packet {NEG-has NEG be.PP / NEG-be.PST.IPF NEG} enough small for
 tenir dans la boîte aux lettres (#mais je l’ai fait entrer).
 fit.INF in the box of-the letters (#but I it-have make.PP enter.INF)
 ‘The packet {was-PFV / was-IPF} not small enough to fit in the mailbox, (#but I
 made-PFV it go in).’

The ostensibly circumstantial construction in (46) is different. Here, we observe the two-way, aspect-sensitive inferential profile first described by Hacquard (2005) (see also 19-22): perfective (46a) shows full (entailed) two-way implicativity, while imperfective (46b) is two-way defeasible.

(46) Aspect-sensitive two-way implicativity:

- a. i. Juno a été assez rapide pour gagner la course, #mais elle n’a pas
 Juno has be.PP enough fast for win.INF the race, #but she NEG-has NEG
 gagné.
 win.PP
 ‘Juno was-PFV fast enough to win the race, #but she did not win-PFV.’
- ii. Juno n’a pas été assez rapide pour gagner la course, #mais elle a
 Juno NEG-has NEG be.PP enough fast for win.INF the race, #but she has
 gagné.
 win.PP.
 ‘Juno was-PFV not fast enough to win the race, #but she won-PFV.’
- b. i. Juno était assez rapide pour gagner la course, mais elle n’a pas
 Juno be.PST.IPF enough fast for win.INF the race, but she NEG-has NEG
 gagné.
 win.PP
 ‘Juno was-IPF fast enough to win the race, but she did not win-PFV.’
- ii. Juno n’était pas assez rapide pour gagner la course, mais elle a
 Juno NEG-be.PST.IPF NEG enough fast for win.INF the race, but she has
 gagné.
 win.PP.
 ‘Juno was-IPF not fast enough to win the race, but she won-PFV.’

These data pose two problems. First, although pragmatic reasoning about sufficiency can invite positive implicative inferences, this is insufficient to predict the non-defeasible inference in (46a-i) (or in 19a, 21a). Secondly, a presupposition of circumstantial necessity predicts the negative entailments in (41b), (45a-ii), and perfective (46a-ii), but should also render imperfective (46b-ii)

incompatible with complement realization (see also 20b, 22b). Since all of the examples in (45)-(46) (and 19-22) appear to be circumstantial, the **inferential profile** puzzle (p.13) remains open: these data show that it cannot be explained by modal flavour or grammatical aspect alone.

3.3 Necessity, sufficiency, and grammatical aspect

As discussed above, two-way lexical implicatives are distinguished from their one-way counterparts by the addition of a sufficiency presupposition: i.e., that the complement is guaranteed when the prerequisite introduced by the matrix clause is satisfied. If *enough* claims' complement inferences are indeed driven by the same semantic components as lexical implicativity, a sufficiency condition must be in play in the interpretation of those constructions which give rise to two-way entailments. Moreover, since the two-way entailment pattern only arises with perfective marking, the data we have seen suggest a link between sufficiency and perfectivity, albeit one which may be modulated by modal flavour or other features of the matrix-complement relationship. We thus have two analytical choices: either the perfective aspect can (in certain contexts) work to introduce a sufficiency condition, or sufficiency must be baked into the semantics of *enough* claims, and mitigated (in some way) in those contexts where positive complement inferences are found to be defeasible.

Hacquard (2005) pursues the second option, treating *enough/assez* predicates as modally-variable counterparts of lexical implicatives. On her proposal (47), *enough/assez* combines with a gradable property Q , a one-place predicate P , and an individual x , establishing that x has at least that degree d of Q presupposed to be necessary and sufficient for complement realization:²⁴

$$(47) \quad \llbracket \text{enough} \rrbracket := \lambda w \lambda P \lambda Q \lambda x. Q(w)(x)(\iota d : \forall w' \in \text{OPT}_{f,g}(w)[Q(w')(x)(d) \leftrightarrow P(w')(x)])$$

This predicts two-way complement entailments when interpreted with respect to a realistic modal domain. The analysis is applied to a perfective circumstantial *assez* claim in (48) adopting (for the moment) Hacquard's treatment of the perfective as existentially quantifying over past times. As with *manage* (compare proposal 15), the combination of presupposition and assertion entails the truth of the complement in (48a) and its falsity in (48b); crucially, these inferences are not dependent on the effects of the perfective, but follow directly from the meaning of *enough/assez*.

- (48) a. $\llbracket \text{Nima a été assez rapide pour s'enfuir} \rrbracket^{w^*, t^*} = 1$ iff
 $\llbracket \text{PFV}(\text{Nima be fast enough to escape}) \rrbracket^{w^*, t^*} = 1$ iff
 $\exists t[t \prec t^* \wedge \text{speed}(w^*)(t)(\mathbf{N}) \geq (\iota d : \forall w \in \text{CR}(w^*)[\text{speed}(w)(\mathbf{J}) \geq d \leftrightarrow \text{escape}(w)(\mathbf{N})])]$
- b. $\llbracket \text{Nima n'a pas été assez rapide pour s'enfuir} \rrbracket^{w^*, t^*} = 1$ iff
 $\llbracket \text{PFV}(\text{NEG}(\text{Nima be fast enough to escape})) \rrbracket^{w^*, t^*} = 1$ iff
 $\exists t[t \prec t^* \wedge \text{speed}(w^*)(t)(\mathbf{N}) < (\iota d : \forall w \in \text{CR}(w^*)[\text{speed}(w)(\mathbf{J}) \geq d \leftrightarrow \text{escape}(w)(\mathbf{N})])]$
- c. *At some time before speech time, Nima had (48a) / lacked (48b) the necessary and sufficient speed for escaping.*

To explain why the two-way entailment pattern is not observed in the imperfective, Hacquard proposes that the French *imparfait* can introduce a layer of modal quantification that counteracts the lexical effects of *enough/assez* (see also Bhatt 1999). More precisely, she suggests that the imperfective can be interpreted as the genericity operator in (49), which requires that a restricted

²⁴Combining necessity and sufficiency allows us to simplify reference to the threshold degree d_n : since all degrees d below d_n are such that lower degrees of the matrix property block complement realization, $\max$ is needed to isolate d_n in the necessity-only case. I thank a reviewer for pointing this out. Once it is also treated as sufficient, the d_n is unique: no degree less than d_n is such that all degrees that meet or exceed it ensure complement realization.

subset of the maximally normal accessible worlds are ones which verify a proposition P . Following Schubert and Pelletier (1989), GEN picks up any relevant restrictions on normality—crucially including any presuppositions of P —via a context-sensitive propositional argument R . The necessity/sufficiency presupposition of *enough* thus gets pushed into the restriction of GEN, deriving the truth conditions in (50) for the imperfective counterparts of (48a)-(48b). Since nothing ensures the normalcy of the evaluation world—nor that it satisfies *enough*’s presuppositions—(50a)-(50b) (like other generic claims) do not require verifying instances: (50a) can be true even if Nima’s actual speed does not meet the escape threshold, and (50b) does not require him to have a below-threshold actual speed, blocking implicative entailments.

- (49) $\llbracket \text{GEN} \rrbracket := \lambda w \lambda R \lambda P. \forall w' \in \text{NORM}(w)[R(w') \rightarrow P(w')]$
- (50) a. $\llbracket \text{Nima \acute{e}tait assez rapide pour s'enfuir} \rrbracket^{w^*} = 1$ iff
 $\llbracket \text{GEN}(\text{Nima be fast enough to escape}) \rrbracket^{w^*} = 1$ iff
 $\forall w \in \text{NORM}(w^*)$
 $[(\exists d : \forall w' \in \text{CR}(w)[\text{speed}(w')(\mathbf{N}) \geq d \leftrightarrow \text{escape}(w')(\mathbf{N})]) \rightarrow \text{speed}(w)(\mathbf{N}) \geq d]$
- b. $\llbracket \text{Nima n'\acute{e}tait pas assez rapide pour s'enfuir} \rrbracket^{w^*} = 1$ iff
 $\llbracket \text{GEN}(\text{NEG}(\text{Nima be fast enough to escape})) \rrbracket^{w^*} = 1$ iff
 $\forall w \in \text{NORM}(w^*)$
 $[(\exists d : \forall w' \in \text{CR}(w)[\text{speed}(w')(\mathbf{N}) \geq d \leftrightarrow \text{escape}(w')(\mathbf{N})]) \rightarrow \text{speed}(w)(\mathbf{N}) < d]$
- c. *All normal worlds where there is a circumstantially necessary and sufficient degree of speed for Nima to escape are worlds where he has (50a) / lacks (50b) that speed.*

The account from genericity can be extended to circumstantial English *enough* claims if we assume that English makes real but covert aspectual distinctions, so that a past-tense *enough* claim is ambiguous between episodicity (perfectivity) and genericity (imperfectivity). If this is right, circumstantial *enough* should be non-implicative just in case it is interpreted generically.²⁵

Hacquard’s analysis predicts that the link between **inference strength** and grammatical aspect should be categorical for circumstantial *enough*: all perfective claims should be two-way implicative, and complement inferences should always be defeasible in the imperfective (as well as in English). This prediction goes against the observed (C pattern) contrasts in (45) (see §3.2, p.20), which permit non-implicative readings for perfective circumstantial *assez* claims. Similarly, while Hacquard uses the construction in (51) to motivate lexicalized sufficiency, arguing that it is only felicitous if Nick’s fame is taken for granted (thereby making his wealth sufficient as well as necessary for an invitation), my consultants diverge. In particular, while they agree that an implicative reading is extremely natural here, they find that Pierre can—without contradiction—continue to state that Nick was in fact not invited to the party because he was insufficiently famous.

- (51) Context. *I am having a conversation with my cousin in L.A. about the glamorous life of Hollywood’s jet-set. She tells me about a very exclusive party on the hills and in the course*

²⁵It is a little tricky to assess this prediction empirically. Claims like (1a)-(1b) sound contradictory to me, but should have an acceptable reading (where only the first clause is interpreted generically) on Hacquard’s analysis. However, one might reasonably argue that this is a pragmatic or processing effect, resulting from the difficulty of assuming that the first clause contains a different covert aspectual operator than the second.

- (1) a. Nima was fast enough to escape, ?/#but he wasn’t (actually) fast enough.
b. Nima was not fast enough to escape, ?/#but he was (actually) fast enough.

of the conversation, she establishes that in order to get invited, one has to be both rich and famous. Her friend Pierre jumps into the conversation and says:

Nick a été assez riche pour se faire inviter.
 Nick has be.PP enough rich for REFL make.INF invite.INF

‘Nick was-PFV rich enough to be invited.’

Default implicativity in (51) can be explained pragmatically—i.e., without treating *être assez riche* (‘be rich enough’) as two-way implicative—along the lines suggested in §3.2. In describing a contingent result, there is a well-known tendency for mentioned conditions to be interpreted *exhaustively* (cf. Geis and Zwicky 1971 on *conditional perfection*, Nadathur and Lauer 2020; Nadathur 2023a on *antiperfection*), which would in this case lead the listener to reason from Pierre’s mention of a single necessary condition to the conclusion that no other necessary condition was contextually relevant for invitation (i.e., that Nick’s fame was not mentioned because it did not affect his chances of an invitation). We assume the satisfaction of a number of other practical, unmentioned, and presumably necessary conditions (e.g., that the party organizers know Nick) for the same reason.²⁶

Lexical sufficiency is also an issue for deontic constructions. Proposal (47) requires a claim like (52a) to background the deontic necessity and sufficiency of a threshold age for alcohol consumption, as shown in (52b). The necessity condition is easily satisfied—many countries establish a minimum drinking age—but sufficiency here is rather puzzling, since it excludes from the deontically optimal set $DN(w^*)$ (i.e., $OPT_{Circ,deon}(w^*)$) any worlds in which Amira is of age but abstains.

- (52) a. Amira a été assez grande pour boire de l’alcool.
 Amira has be.PP enough old for drink.INF of the-alcohol.
 ‘Amira was-PFV old enough to drink alcohol.’
 b. $\llbracket (52a) \rrbracket^{w^*,t^*} = \llbracket PFV(\text{Amira be old enough to drink alcohol}) \rrbracket^{w^*,t^*} = 1$ iff
 $\exists t[t \prec t^* \wedge \text{age}(w^*)(t)(A) \geq (\lambda d : \forall w \in DN(w^*)[\text{age}(w)(A) \geq d \leftrightarrow \text{drink}(w)(A)])]$
There is a past time at which Amira’s actual age was at least the age which is deontically necessary and sufficient for her to drink alcohol.

Absent laws mandating alcohol consumption by those who are at least the minimum drinking age, a sufficiency presupposition can only be enforced here via the modal base: i.e., we must assume that Amira’s circumstances (and dispositions) are such that achieving the age of majority guarantees that she will drink. As discussed in §3.2, this is precisely the sort of context in which satisfying a deontically necessary condition can be construed as a trigger for (attempted) complement realization, predicting an implicative interpretation for (52a). The upshot of lexicalizing sufficiency, then, is that (52a) should only be felicitous in contexts where Amira’s drinking is assured, a result that is challenged by the C judgement for (43a) and the C and R judgements for (44) (p.19).

An additional argument for treating a potential sufficiency condition differently than necessity comes from judgements of (ostensibly) imperfective and circumstantial *enough* and *assez* claims. If sufficiency- and necessity-driven inferences are placed on the same lexical footing (as in 47), and if it is also the case that the sufficiency-driven inferences can be abrogated by a (covert) genericity operator, as we seen in imperfective data like (45a-i) and (45b-i) (p.20) and the corresponding

²⁶ A non-implicative reading for (51) is also possible if Nick’s fame is taken for granted, as long as it is understood that the time at which invitations were issued is after the reference time (at which Nick’s wealth is assessed): on this reading, (51) conveys that Nick’s reference time wealth met the invitation threshold, but he has since become poorer, precluding the *assez* complement at invitation time. This reading seems available on both the C and R patterns.

English claims (40a) and (41a) (p.17), then we should find the same for necessity-driven inferences. This seems to be true in certain cases, as shown in (46b) (p.21), but my consultants reject the cancellation of implicative inferences in (45a-ii) and (45b-ii), regardless of aspectual marking, and these judgements are replicated for the corresponding *enough* constructions in (40b) and (41b).

Although the C and R patterns do not agree on all of the data cited here, both patterns include judgements that go against treating a sufficiency condition as a lexical presupposition of *enough/assez*. Contra Hacquard, then, this suggests a line of analysis on which *enough* predicates lexicalize only a necessity presupposition (as proposed in §3.1), but are subject to enrichments that strengthen this condition to one of joint necessity and sufficiency under certain grammatical and/or pragmatic pressures. From this perspective, the questions to be examined are as follows: how and why can perfective aspect privilege or even mandate this enrichment in a subset of cases, and what other features of the relevant constructions play a role in the effect?

3.4 Aspectual coercion, sufficiency, and adjective type

The structure of the approach I am proposing is the same whether we attempt to explain the C or R judgement pattern: my claim is that the perfective aspect can (under certain conditions) affect the interpretation of a necessity-only *enough* predicate in such a way that a sufficiency condition (specifically, an enrichment of necessity to a condition of practical necessity and sufficiency) becomes both relevant and ‘active’, producing implicative entailments. This approach is based on **aspectual coercion** (see, a.o., Moens and Steedman, 1988; de Swart, 1998; Bary, 2009).

The core idea behind aspectual coercion is that aspectual operators—including grammatical aspects—can have *selectional restrictions*, composing only with eventuality predicates bearing certain aspectual class features (such as *quantization*, *homogeneity*, or *eventivity*; Krifka 1998; de Swart 1998). Mismatches between an operator and an input predicate can be repaired (and composition achieved) if the predicate is reinterpreted as a predicate of closely related eventualities with the right formal properties. The process of reinterpretation is called *coercion*, and does not occur freely: it acts as a type of last resort which must be triggered by a mismatch, and is further constrained by selection from a small (and potentially language-specific) set of coercion operators, one of which must provide a coherent and plausible reinterpretation of the input predicate in the context at hand. I discuss some commonly-noted forms of coercion in §4.2. My suggestion here is that composing a perfective marker with an *enough* predicate involves an aspectual mismatch—supported by the observation that perfectivity affects event structure (see, e.g., the discussion of 43-44 on p.19)—and that, in a specific subset of cases, the repair for this mismatch results in the introduction of the ‘missing’ sufficiency condition.

The C and R patterns of interpretation differ with respect to which predicate-complement combinations are subject to sufficiency-inducing aspectual coercion. In building towards a characterization of the licensing features, I focus on the C pattern for French, which appears to be somewhat more selective, and with which my English judgements align. §4 takes up a formal treatment of coercion in *enough* constructions, and I there introduce an operator which will derive the C pattern as well as an alternative—drawn from Homer’s (2021) coercion-based treatment of *actuality entailments* (Bhatt, 1999)—that seems to align with the R judgements.

If sufficiency-inducing coercion is to apply to ‘two-way’ circumstantial constructions like (46), but not to their one-way alternatives (e.g., 45), there must be some systematic semantic difference between the aspectually modified predicates in these two classes. I suggest that this has to do with the nature of the property invoked by the matrix adjective. The properties attributed by one-way

constructions differ from those of two-way constructions in the way that they can be embodied or manifested by an individual, with the result that while ‘one-way’ properties act as *enabling* conditions for *enough*’s complement, only ‘two-way’ properties introduce precipitating *causes*.

The one-way implicativity profile arises from circumstantial constructions with matrix adjectives that pick out **static** properties: these are adjectives such as *tall*, *small*, or *old* that describe attributes with no inherent relationship to action. Static adjectives can be contrasted with **dispositional** adjectives (e.g., *clever*, *fast*, *polite*; see, a.o., Krifka et al. 1995; Piñón 2015; Demonte 2019) that characterize individuals in view of propensities that can only be measured (made manifest) in actions: a dispositional matrix predicate is a necessary (albeit not sufficient) condition for the two-way implicativity profile.²⁷ Why the static/dispositional contrast matters will be illuminated by examining the reasoning involved in linking an *enough* property attribution to the realization of a particular complement eventuality.

Let us consider the one-way circumstantial construction in (53). It should be intuitively clear why having the necessary height (d_n) for branch-grasping will not in general be sufficient to produce this result. Having height d_n will certainly enable Nima to reach the branch should he be in a position to do so, but actually grasping it depends on a number of other causing and enabling conditions, such as his physical location, his range of motion, and—most importantly—whether or not he actually moves his arm in an attempt to grasp the branch. It is only when all of these other conditions are met that Nima’s height can ensure complement realization.

(53) Nima was tall enough to grasp the branch.

Another way of thinking about this is to observe that calculating the threshold value d_n for a given *enough* construction inherently involves reasoning about *contingent* or *hypothetical* sufficiency. In (53), we collect a set of values for Nima’s height across optimal branch-grasping worlds: if these worlds are normal (non-magical), they must be ones in which any non-height conditions for branch-grasping are satisfied. The threshold degree d_n extracted from this set thus corresponds to the minimum height that ensures a branch-grasping outcome in contexts where it is the only necessary condition in question. As discussed in §3.2, contingent sufficiency can become pragmatically ‘active’ if we assume that the evaluation world is one in which all other preconditions for branch-grasping are satisfied, but (53) will not give rise to a complement entailment unless we independently assert that these other conditions are positively resolved.

Contingent sufficiency is also relevant for dispositional constructions like (54): determining d_n here requires considering Juno’s speed in normal worlds where she wins the race. These worlds must be ones in which a number of other conditions are satisfied: Juno must be uninjured, she must show up at the start line, and—most importantly—she must make an attempt to win, presumably by starting to run once the gun sounds. The truth of (54) amounts to the assessment that Juno’s actual speed—that is, the speed of which she is capable—suffices for her to win the race just in case all other practical preconditions are satisfied, and she actually tries to win.

(54) Juno was fast enough to win the race.

²⁷The dispositional class includes both *mental propensity* adjectives like *generous*, *polite*, and *stupid* (often termed *evaluative*), which have received a fair amount of attention (see, a.o. Wilkinson 1976; Stowell 1991; Kertz 2006; Fábregas et al. 2013; Martin 2015) based on their resistance to classification as either individual- or stage-level predicates (Kratzer, 1995), as well as *physical propensity* predicates (Dixon, 1982) like *fast*, *nimble*, and *loud*. What these classes have in common is that they are understood to characterize a propensity or capacity to engage in actions which can be assessed (whether on a subjective/evaluative or an objective basis) in terms of the relevant property.

It is here that (53) and (54) come apart. In both examples, the proximate cause of complement realization is the subject’s attempt to do it. The success of an attempt in (53) is facilitated by height, but is in some sense conceptually independent of it: the process of Nima moving his arm upwards is not in any way characterized by height. The same cannot be said for (54). Here, Juno’s capacity for speed cannot be disentangled from an attempt at race-winning, since the action which constitutes the (potential) proximate cause of winning is necessarily characterized by speed: put another way, the attempt must constitute a *manifestation* of Juno’s speed. Contingent sufficiency in (54) thus amounts to a condition of *causal sufficiency* between Juno’s (race-time) manifestation of speed d_n establishing that Juno’s (race-time) manifestation of speed d_n and the event of her winning the race. My claim is that two-way constructions are differentiated from their one-way counterparts by the relevance of this condition, which in turn relies on two things: the dispositional nature of the matrix adjective, and the characterizing relationship that this property bears to the proximate cause of the complement.

I propose that one-way claims represent a class of ‘true’ (contextually unenriched) circumstantial *enough* constructions, subject only to the necessity condition formalized in §3.1. Two-way constructions license a naturally causal view of the relationship between the matrix property and the complement. These constructions share the necessity semantics—linking a stative (i.e., latent) capacity to their complements—but are automatically enriched with a causal background structure supporting the joint causal necessity and sufficiency of a manifestation of the matrix property for complement realization. This condition can be formalized as in (55), but I do not claim that dispositional predicates are semantically more complex than other *enough* predicates: rather, (55) is simply a consequence of combining the necessity semantics with world knowledge pertaining to the relationship of a particular disposition and the goal specified by the *enough* complement.

- (55) **Causal necessity/sufficiency** for $\llbracket \text{enough}(P)(Q)(x) \rrbracket^w$, where x is an individual, P is a one-place predicate, and Q a disposition:

$$\forall w' \in \text{CS}(w) [\text{INST}(Q(w')(x)(d_n)) \leftrightarrow P(w')(x)]$$

where $d_n = \max(\{d : \forall w' \in \text{OPT}_{\text{cr}, \emptyset}(w) [P(w')(x) \rightarrow Q(w')(x)(d)]\})$ and $\text{INST}(Q(w')(x)(d_n))$ is a manifestation to degree d_n of Q by x

That (55) is relevant for dispositional *enough* constructions does not by itself predict a two-way pattern of complement entailments, which is as desired. It is of course possible to attribute to the *enough* subject a (capacity for the) complement-realizing degree of the matrix property without entailing that this property is manifested at reference time. This is where the effects of temporal and aspectual modification come into play: as discussed in the next section, perfective-induced aspectual coercion is one way to force the shift from a stative disposition to an eventive manifestation. A reinterpretation of this sort can happen only with dispositional (actionable) properties, and even then will only result in implicative entailments in case world knowledge supports a proximate-cause relationship between the manifested disposition and the *enough* complement.

The required causal relationship further restricts the distribution of two-way implicative entailments even within the class of circumstantial and dispositional *enough* claims. The two-way pattern requires a matrix property that plausibly characterizes the proximate cause of the complement; moreover, the reference time of the matrix clause (interpreted as an action) should overlap or at least directly abut the time of the complement, so that a matrix manifestation of the (necessary) adjectival degree can in fact be identified with a good faith attempt at complement realization.

Both of these predictions appear to be upheld.

Example (56a) is informative with respect to the relationship between a disposition and an *enough* complement. While some degree of strength is presumably necessary for winning any race, strength is not usually the defining characteristic of a win attempt. Accordingly, (56a) only invites implicative interpretations if we supply a specialized context in which strength rather than speed is indeed determinative: suppose for instance that Juno is competing in an endurance event, which one wins not by crossing the finish line first, but by completing the course distance while carrying the most weight. In a more typical context, however, (56a) might be a natural claim to make if Juno is known to be recovering from illness, in order to assert that her current strength will not be an obstacle; in this case, speed is still understood to be the actual proximate cause of a potential win, and (56a) is only implicative if we independently supply the remaining necessary conditions (including that Juno was capable of the necessary speed, and in a position to manifest it). An example like (56b), in which the dispositional matrix property plausibly characterizes the process of goal-realization, licenses implicativity with minimal context.

- (56) a. Juno was strong enough to win the race. *context-dependent*
 b. γ [Swinkels] was nimble enough to scoop it up before the ball crossed the goal line.
 $\leadsto$ *Swinkels scooped up the ball in time.*

A temporal constraint on implicativity is supported by the English and French data in (57a)-(57b). It is felicitous in both languages to temporally separate the matrix and complement clauses, but this disrupts implicativity, even in the presence of overt perfectivity: on the current analysis, this is because manifesting the causing property for complement realization cannot actually bring about the complement if it happens at some other time or place. Additional corroboration comes from Marques (2012), who independently proposes a temporal constraint on *enough* implicativity in Portuguese: his example (57c) is not complement-entailing, despite perfective marking.²⁸

- (57) a. Quand je l'ai vue courir la semaine dernière, Juno a été assez
 When I she-have see.PP run.INF the week last, Juno has be.PP enough
 rapide pour gagner la course d'aujourd'hui, mais il s'est avéré qu'elle
 fast for win.INF the race of-today, but it REFL-is establish.PP that-she
 a couru beaucoup plus lentement aujourd'hui.
 has run.PP much more slow today.
 'When I saw-PFV her run last week, Juno was-PFV fast enough to win today's race, but
 it turned out that she ran-PFV much more slowly today.'
- b. When I watched her run last week, Juno was fast enough to win today's race.
compatible with: Juno ran too slowly to win today; Juno did not participate, etc.
- c. No último encontro, ele foi humilhado o suficiente para agora recusar o
 in.the last meeting, he was.PFV humiliated the enough to now refuse the
 convite para um novo encontro (mas parece que já se esqueceu, porque
 invitation for a new meeting (but appears that already REFL forgot since
 está a pensar aceitar).
 is at think accept.INF).

²⁸Marques (2012) additionally argues that implicative readings for Portuguese *enough* claims arise only with evaluative (mental propensity) adjectives such as *corajosa* ('brave') and *inteligente* ('intelligent'). He suggests that this has to do with how such properties are measured—in particular, that they can only be measured by way of the process of realizing the *enough* complement—which is compatible with the dispositional-predicate account I propose here.

'In the last meeting, he was humiliated enough to refuse now the invitation for a new meeting (but apparently he already forgot, since he is planning accept the invitation).'

Taken together, these data provide strong support for the view that implicative interpretations for *enough* claims depend on an interpretation of the matrix property attribution (i.e., the core assertion) as describing not a latent disposition, but instead a complement-time realization of what is backgrounded as a causally necessary and sufficient condition for complement realization.

The discussion above restricts two-way implicative entailments to circumstantial *enough* constructions with dispositional adjectives and the appropriate temporal specifications. Insofar as condition (55) is inherently relevant in these cases, it also suggests that these constructions are likely to generate pragmatic but defeasible inferences to implicativity even where no grammatical device forces the manifested reading of the matrix property attribution. My claim is not that other *enough* constructions cannot be interpreted implicatively, but rather that specific local context is needed to supply the practical-sufficiency relationship between satisfying the matrix condition and realizing the complement. Thus, as discussed previously, one way constructions and even certain deontic constructions can also give rise to (defeasibly) implicative interpretations as long as their utterance contexts license the assumption that any preconditions for complement realization other than the one specified by the *enough* matrix clause are met.

3.5 Capacity and manifestation: the role of grammatical aspect

The proposal is that a dispositional and circumstantial *enough* claim with the appropriate causal background and temporal specifications will license an implicative entailment when the matrix property attribution is interpreted as a manifesting action. The two interpretive possibilities—latent capacity and manifestation—correspond to a difference in what an *enough* claim asserts. A capacity attribution establishes the possibility that the subject will act on the proximate cause of the complement, but cannot guarantee that this occurs. An eventive (manifestation) interpretation, on the other hand, is one on which sufficient cause for the complement is realized: this ‘activates’ the causal sufficiency condition in (55) and licenses implicativity. On the eventive reading, then, a dispositional circumstantial claim shares both the causal background structure and the asserted content of a lexical implicative: the two-way implicative profile should thus co-occur with instantiated interpretations of the matrix attribution.

This seems to be the case. English dispositional claims are systematically ambiguous between stative (capacity-attributing) readings and eventive (manifesting) readings:

- (58) Juno was fast. *Eventive*: At some past time, Juno did something quickly.
Stative: At some past time, Juno had the capacity for quick action.

The two readings can be disambiguated by temporal modification affecting the length of the reference interval (and, by proxy, the episodic nature of the main claim). Short reference intervals, as in (59a), license manifesting readings more readily than extended reference periods (as in 59b).

- (59) a. Juno was fast this morning. *Eventive*: She did something fast this morning.
b. In her twenties, Juno was fast. *Stative*: She had the capacity to do fast things.

The same pattern extends to *enough* claims: the short, episodic context in (60a) most naturally picks out an event of Juno doing something at speed d_n or higher. Coupled with the causal presupposition in (55), this licenses the inference that she won the race. The preference for a

manifesting reading is strong enough that—absent further context—complement cancellation is extremely marked. (60b), on the other hand, privileges a stative reading, and is thus perfectly compatible with complement denial: where we have no reason to assume that sufficient cause for the complement occurred, we also have no reason to infer that the complement was realized.

- (60) a. At today’s track meet, Juno was fast enough to win her race, ??but she did not win.
 b. In her twenties, Juno was fast enough to win local 5Ks, but she never participated.

In French, grammatical aspect disambiguates the two interpretations as predicted by a coercion-based approach (cf. Homer 2011). Perfective (61a) describes a manifestation of speed, while its imperfective counterpart in (61b) licenses the pure capacity reading. The result is that perfective dispositional *assez* claims can be non-defeasibly implicative: since the matrix clause in (46a-i) is interpreted as a manifestation of the necessary/sufficient cause of complement realization, the complement itself follows via causal entailment. Imperfective *assez* in (46b-i) remains compatible with both complement realization and complement failure.

- (61) a. Juno a été rapide (#mais elle n’a rien fait).
 Juno has be.PP fast (#but she NEG-has nothing do.PP).
 ‘Juno was-PFV fast (#but she didn’t do-PFV anything).’
Eventive: Juno did something fast.
 b. Juno était rapide (mais elle n’a rien fait).
 Juno be.PST.IPF fast (but she NEG-has nothing do.PP).
 ‘Juno was-IPF fast (but she didn’t do-PFV anything).’
Stative: Juno was capable of doing fast things.
- (46a-i) Juno a été assez rapide pour gagner la course, #mais elle n’a pas gagné.
 Juno has be.PP enough fast for win.INF the race, #but she NEG-has NEG win.PP
 ‘Juno was-PFV fast enough to win the race, #but she did not win-PFV.’
- (46b-i) Juno était assez rapide pour gagner la course, mais elle n’a pas gagné.
 Juno be.PST.IPF enough fast for win.INF the race, but she NEG-has NEG win.PP
 ‘Juno was-IPF fast enough to win the race, but she did not win-PFV.’

The puzzle of inference strength is thus a consequence of the aspectual class ambiguity inherent to dispositional adjectives. Since disambiguation is largely contextual for English, nothing forces a particular claim to take a manifested interpretation, leading to a defeasible tendency for complement inferences from dispositional (causal) *enough* claims. In French, however, interpretation is constrained by grammatical aspect. Perfectivity privileges a manifested reading for dispositional *assez*, so a speaker will choose the imperfective if the stative reading is intended.

It is worth pointing out that the coercive interaction between perfective marking and dispositions is observed independently of the other features required for *enough* implicativity. We observe the manifested reading for the disposition attribution in both (62a) and (62b)—both examples describe an event of Juno running at or above the necessary speed for complement realization—but only (62a) licenses an implicative entailment. This is because only (62a) supports the required proximate-cause relationship between matrix and complement. Running a sufficiently fast time (in an appropriate qualifying race) is an immediate and sufficient cause of qualifying for the Boston Marathon (qualifying happens when the qualifying time is run), but only acts as a deontic precondition for actually signing up for Boston. The comparison between these two examples shows that

the effect of adjective type must conspire with aspect and modal flavour to produce implicative behaviour in an *enough* construction, as suggested in §3.2.²⁹

- (62) a. Juno a été assez rapide pour se qualifier pour courir le marathon de
 Juno has be.PP enough fast for REFL qualify.INF for run.INF the marathon of
 Boston, #mais elle ne s'est pas qualifiée.
 Boston, #but she NEG REFL-is NEG qualify.PP
 'Juno was-PFV fast enough to qualify for the Boston Marathon, #but she did not
 qualify.'
- b. Juno a été assez rapide pour s'inscrire au marathon de Boston,
 Juno has be.PP enough fast for REFL-register.INF at.the marathon of Boston,
 mais elle ne s'est pas inscrite.
 but she NEG REFL-is NEG register.PP
 'Juno was-PFV fast enough to register for the Boston Marathon, #but she did not
 register.'

We are now quite close to a full picture of *enough* implicativity, having explained both the **optionality** and **inference strength** puzzles. We have also linked the puzzle of **inference profile** variation to the nature of a circumstantial construction's matrix property, but some loose ends remain. Most immediately, we still need to explain why, despite an across-the-board necessity condition, the complement inferences of two-way imperfective *assez* constructions (and, to a lesser extent, those of *enough*) can be cancelled under negation, while those of one-way constructions in the same grammatical context are not (compare, for instance, 22b, 46b-ii to 23b).

This effect is also tied to the contrast between static and dispositional properties. As it turns out, dispositions can take both stative and eventive (*progressive*) interpretations under imperfective marking. The eventive reading is prominent in (63a): here the matrix clause describes an in-progress event of Juno running slower than the race-winning threshold d_n , but does not commit its speaker to the normal (predicted) result of this performance—i.e., that Juno does not win the race. While progressive readings are not prominent for aspectually-unmarked English dispositions, (63b) is naturally read in parallel with (63a).

- (63) a. (Au début,) Juno n'était pas assez rapide pour gagner la course,
 (at-the start,) Juno NEG-be.PST.IPF NEG enough fast for win.INF the race,
 mais dans le dernier tour, elle a couru beaucoup plus vite et a finalement
 but in the final lap, she has run.PP much more quickly and has finally
 réussi à gagner.
 succeed.PP at win.INF.
 '(At the start of the race,) Juno was-IPF not (running) fast enough to win, but in the
 final lap she ran-PFV much faster and ultimately managed-PFV to win.'
- b. (At the start,) Juno was not fast enough to win the race, but she sped up so much on
 the last few laps that she managed to win after all.

²⁹There is an implicative reading of (62b) (on which the non-implicative continuation becomes infelicitous): on this reading, the matrix clause describes Juno's speed in the registration process, rather than in a qualifying race. Rather tellingly, however, this reading loses its deontic flavour and replaces it with the causal background structure in (55): Juno's speed in the registration process (rather than the race) is the proximate cause for her successful registration.

Non-implicativity in (63a)-(63b) exemplifies the well-known *imperfective paradox* (Dowty, 1979), in which progressives of goal-oriented events can be true even if the described event-in-progress goes awry or ends prior to its expected point of culmination. In (63), the (causally) predicted result of manifesting insufficient speed is that Juno loses the race; however, since in-progress readings zero in on some point within the race, the matrix claim in both examples is compatible with her subsequently (after reference time, but while the race is still developing) speeding up and thereby eking out a win.³⁰ Static predicates resist progressive interpretations: English (64a) does not describe Nima’s performance of his inadequate height, and my consultants likewise report that the negated *assez* claim in (64b) resists an in-progress reading.

- (64) a. Nima was not tall enough to grasp the branch.
Impossible: #Nima was not being tall enough to grasp the branch.
- b. Nima n’était pas assez grand pour attraper la branche, #mais il l’a
 Nima NEG-be.PST.IPF NEG enough tall for grasp.INF the branch, #but he it-has
 fait.
 do.PP.
 ‘Nima was-IPF not tall enough to grasp the branch, #but he did-PFV it.’
Impossible: ‘#Nima was not being tall enough to grasp the branch.’

As it turns out, negated two-way *enough* constructions can also be defeasibly implicative on the stative (capacity-attributing) reading. In these interpretations, the main clause describes what is typical for the individual in question (over the reference period), while the complement-affirming continuation pinpoints a specific context in which a manifestation of the matrix property temporarily deviates from the norm. Dispositions incorporate an inherent exception tolerance, so that it is possible to truthfully attribute the capacity for less-than-race-winning speeds to Juno over a reference interval while observing a singular performance of higher speed. This works only insofar as the race-winning performance is indeed regarded as exceptional or aberrant—i.e., as long as we assume that the strong majority of Juno’s other speed performances during the reference interval conform to the (slower) norm. Readings of this sort are readily attested in French, but are marginal in English in the absence of an overt marker of habituality or typicality; one reason for this difference may be that—in the absence of overt aspectual marking in English—additional context is needed to ensure the imperfective reading of the *enough* claim.

- (65) a. Juno was not ??/ordinarily fast enough to win the race, but (due to a drug) she ran
 extraordinarily quickly that day and actually managed to win.
- b. Juno n’était pas assez rapide pour gagner la course, mais (à cause
 Juno NEG-be.PST.IPF NEG enough fast for win.INF the race, but (from cause
 d’un médicament) elle a couru inhabituellement vite à cette fois-là et
 of-a drug) she has run.PP unusually quickly on DEM time-DIST and
 elle a réussi à gagner.
 she has succeed.PP at win.INF.
 ‘Juno was-IPF not fast enough to win the race, but (because of a drug) she ran unusually
 fast on that occasion and managed to win.’

³⁰Progressive readings are also possible for positive *enough* and *assez* claims (with imperfective marking), and these cases are also non-implicative; see §4.4.

Static properties do not tolerate exceptions. The attribution of a property like *tall* or *old* to an individual constitutes a fixed measurement: one’s height or age may change gradually over an extended period, but does not admit temporary fluctuation. If one has height below d_n in some reference period, it is not possible for there to be moments within this timeframe at which one is briefly d_n -*tall*, and thus there can be no points within the reference period at which the subject of a negated static claim actually verifies the necessity condition associated with *enough/assez*. A claim like (66) is thus only acceptable in a fantastical context in which individual hands and/or gloves can fluctuate in size, but is ruled out in any realistic context.

- (66) Alice’s hand was not (??)ordinarily small enough to fit into the glove, but (due to a magical temporary change in size) it fit on that occasion.

Both of the inferential contrasts which distinguish two-way from one-way circumstantial *enough* claims—full complement entailment in positive, perfective cases and defeasible complement denial for negated imperfective (and some English) claims—are thus linked to the nature of the matrix adjective. One-way claims involve static properties that are stable over time, attributed directly to an individual, and which can enable realization of the *enough* complement, but do not characterize its proximate cause. Two-way claims license the aspect-sensitive profile first observed by Hacquard (2005) because they involve dispositional properties that are causally involved in complement realization, and which (insofar as they can only be directly measured via manifestation) allow both eventive and stative readings, the latter of which are compatible with temporary deviations.

3.6 Summary

Table 1 summarizes the parameters of variation in the implicativity of *enough* constructions. Blanks in the table indicate that a given parameter does not affect inference licensing; $+/-$ in the rightmost columns indicate matrix polarity (as well as the polarity of the associated complement inference). Optional implicativity is tied to modal flavour, variation in inferential profile to a contrast in properties of the matrix adjective, and inference strength variation results from a complex, aspect-governed interaction between adjective type, aspectual class, and temporal interpretation. In order to be fully implicative, *enough* constructions must be circumstantially interpreted, involve a dispositional adjective which characterizes the necessary/sufficient proximate cause of complement realization, and provide overlapping temporal specifications for matrix and complement clauses; given these features, implicative entailments arise where the matrix property attribution is interpreted not as description of latent capacity, but instead as a manifestation of the complement-causing action.

Table 1 makes good on the thesis (cf. Hacquard 2005) that *enough* complement inferences are genuinely implicative. These inferences are derived from the same semantic components as the complement entailments of lexical implicatives: (i) the backgrounded relevance of a necessary/sufficient prerequisite for complement realization, (ii) a causal interpretation of necessity and sufficiency, and (iii) an assertion which settles whether or not the causal prerequisite occurs. These components are all lexically encoded by implicative verbs (see proposal 13), but emerge through a complex set of contextual and compositional interactions for *enough* constructions.

Enough claims lexically impose only a modally-variable necessity relationship between their matrix properties and their complements. However, when a circumstantial *enough* construction contains a (suitable) dispositional predicate, the lexical semantics of *enough/assez* conspire with world knowledge to induce a background condition that replicates the causal background structure of lexical implicatives: an appropriately-timed manifestation of the matrix property is taken to be

<i>modal flavour</i>	parameter of variation			inference	
	<i>adjective type</i>	<i>temporal spec.</i>	<i>aspect</i>	+	−
non-circumstantial (deontic)	—	—	—	$\nrightarrow$	$\nrightarrow$
circumstantial	static	—	—	$\nrightarrow$	$\rightarrow$
		no overlap	—	$\nrightarrow$	$\leadsto$
			unmarked	$\leadsto$	$\leadsto$
	dispositional	overlap	imperfective	$\leadsto$	$\leadsto$
			perfective	$\rightarrow$	$\rightarrow$

Table 1: Governing parameters of *enough* implicativity (C pattern)

causally necessary and sufficient for complement realization. However, while lexical implicatives assert prerequisite realization, dispositional *enough* constructions are compatible with the weaker claim that their subjects are merely capable of realizing the complement’s prerequisite cause. It is only where aspectual marking (or aspectually-relevant context) licenses a manifested interpretation of the property attribution that an *enough* assertion exactly matches that of a lexical implicative. Thus, fully-implicative behaviour arises only where modality, adjective type, and temporal context conspire to recreate implicative semantic structure.

Once these criteria are established, the last piece of the empirical puzzle falls into place. As noted in §1, one of the challenges in explaining *enough* complement inferences along implicatives lines is that they show a pattern of aspect-sensitivity which is not observed for true (lexical) implicatives. Hacquard’s (2005) proposal—on which *enough* claims are lexically two-way implicatives—cannot explain, for instance, the contrast between (67a) and (67b): the same contrast arises between imperfective French *assez* and *réussir* (‘succeed’), provided that the main claim is interpreted non-progressively. The current account offers a straightforward explanation for these contrasts.

- (67) a. Juno was fast enough to win races, but she didn’t win any races.
b. Juno managed to win races, #but she didn’t win any races.
- (68) a. Juno était assez rapide pour gagner les courses, mais elle n’a jamais
Juno be.PST.IPF enough fast for win.INF the races, but she NEG-has never
gagné de course.
win.PP a race.
‘Juno was-IPF fast enough to win races, but she never won a race.’
b. Juno réussissait à gagner les courses, #mais elle n’a jamais gagné de
Juno succeed.PST.IPF at win.INF the races, #but she NEG-has never win.PP a
course.
race.
‘Juno managed-IPF to win races, #but she never won a race.’

As emphasized above, the crucial difference between lexical implicatives and (two-way) implicative *enough* claims is in asserted content. Lexical implicatives assert the realization of their causal

prerequisites: since the prerequisite in question is encoded as an action of the implicative subject, this content is at-base eventive. On a non-progressive reading, an imperfectly-marked lexical implicative claim like (68b) is interpreted *habitually*, describing regular or repeated events in which the full implicative claim (both cause and result) is verified. A claim of this sort is compatible with occasional instances of failure, but if complement non-realization is the norm within the reference interval, the event structure outlined by the implicative construction cannot coherently constitute a habit. By contrast, implicative *enough* claims have a ‘true’ stative reading, on which it is established that the sentential subject has the capacity for performing the proximate cause of the complement, but does not require them to act on this capacity. As a result, (68a) is compatible with outright denial of the complement (as well as with periodic or singular failures). The empirical contrasts in (67) and (68), then, fall out from the proposed contrast in the lexical aspectual properties of the two matrix predicates.³¹

This section focused on linking the empirical patterns of variable implicativity to the components and compositional features of *enough* claims. In so doing, I have provided strong evidence in support of the claim that the (aspect-sensitive) complement inferences of *enough* constructions are genuine instances of implicativity. This account relies on the temporal and aspectual properties of *enough* claims, which we have so far discussed only informally: the next section takes up the details of temporal interpretation.

4 Aspect and implicativity

I assume a three-times system (Reichenbach, 1947), on which grammatical aspects relate situation times to reference times, and pronominal tenses (Partee, 1973; Kratzer, 1998) relate reference times to utterance times. I couple this with an event semantics that distinguishes *stative* from *eventive* predicates. Statives describe (Vendlerian) *states*: situations which are *fully divisive* (satisfying a strong form of the subinterval property; Krifka 1998), and which *hold* over temporal intervals, but do not *happen*. Eventives describe *events* (including *processes*; Mouratelos 1978; Bach 1986; Krifka 1998) which *happen*, have potentially varied internal structure and/or involve change. The domain of times $D_i = \langle T, \prec \rangle$ is a partially ordered set of non-empty intervals; two times *overlap* ($t_1 \circ t_2$) iff they cannot be ordered.³² Events (D_v) are mapped to times using the *temporal trace* function τ . I treat eventives as predicates of events (type $\langle v, t \rangle$) and statives as expressing relations between individuals and times ($\langle e, \langle i, t \rangle \rangle$). Assuming that individual arguments are merged below aspect, but that tense applies above aspect, statives will combine with aspects as predicates of times ($\langle i, t \rangle$).

Lexical entries for tenses and aspects are given in (69)-(70) (cf. Klein, 1994; Kratzer, 1998). I begin by assuming that aspects are compatible with both statives and eventives, and simply contrast in how they relate reference and situation times; perfectives contain a situation within reference time (IN), while imperfectives contain the reference time within a situation (ON).

³¹The contrast between imperfective *être assez rapide* (‘be fast enough’) and *réussir* (‘manage’) falls away under a progressive interpretation of both claims, which is also as predicted by the current account. On a progressive interpretation, the *assez* predicate describes an in-progress manifestation of the threshold speed, but—as with progressives of other goal-oriented predicates—this can be interrupted before it leads to complement realization, blocking implicativity. An eventive *réussir* predicate has the same interpretive possibility, describing an in-progress realization of the matrix causing event which can be interrupted before it leads to a complement entailment.

³²The precedence relation $\prec$ is induced via a total order $\prec$ on the set $T_p \subseteq T$ of *temporal points* (comprising minimal elements of T); $t_1 \prec t_2$ iff $\forall t_p \in T_p \cap t, \forall t'_p \in T_p \cap t',$ we have $t_p \prec t'_p$ (see Condoravdi 2010).

- (69) a. $\llbracket \text{PST} \rrbracket^w := t$, defined iff $t \prec \text{now}$ b. $\llbracket \text{PRS} \rrbracket^w := t$, defined iff $t \circ \text{now}$
- (70) a. $\llbracket \text{PFV} \rrbracket^{w,t} := \lambda P. \text{IN}(t, w, P)$, where $\text{IN}(t, w, P) = \begin{cases} \exists e[\tau(e) \subseteq t \wedge P(w)(e)] & \text{for } P \in D_{vt} \\ \exists t'[t' \subseteq t \wedge P(w)(t')] & \text{for } P \in D_{it} \end{cases}$
- b. $\llbracket \text{IPF} \rrbracket^{w,t} := \lambda P. \text{ON}(t, w, P)$, where $\text{ON}(t, w, P) = \begin{cases} \exists e[\tau(e) \supset t \wedge P(w)(e)] & \text{for } P \in D_{vt} \\ \exists t'[t' \supset t \wedge P(w)(t')] & \text{for } P \in D_{it} \end{cases}$

(71) illustrates the interpretation of an eventive past perfective: (71b) requires the existence of an evaluation world event of Maria swimming, contained within a salient past time.

- (71) a. Maria a nagé.
Maria has swim.PP
'Maria swam-PFV.'
- b. $\llbracket (71a) \rrbracket^{w^*, t^*} = \llbracket \text{PST}(\text{PFV}(\sqrt{\text{Maria swim}})) \rrbracket^{w^*, t^*}$
 $= \text{IN}(t^* \{ \prec \text{now} \}, w^*, \lambda w \lambda e. \text{swim}(w)(e)(\mathbf{M}))$
 $= \exists e[\tau(e) \subseteq t^* \{ \prec \text{now} \} \wedge \text{swim}(w^*)(e)(\mathbf{M})]^{33}$

4.1 Temporal semantics for *enough* constructions

Our first task is to update the interpretation of measure statements. I take these attributions to be stative, intuitively describing measurements of individuals that hold at world-time pairs. I apply the updated lexical entry for *old* to an imperfective French claim in (73).

- (72) $\llbracket \text{old} \rrbracket = \lambda w \lambda t \lambda d \lambda x. \text{age}(w)(t)(x) \geq d$
- (73) a. Amira avait 18 ans.
Amira have.PST.IPF 18 years.
'Amira was 18 years old.'
- b. $\llbracket (73a) \rrbracket^{w^*, t^*} = \llbracket \text{PST}(\text{IPF}(\sqrt{\text{Amira be 18}})) \rrbracket^{w^*, t^*}$
 $= \text{ON}(t^* \{ \prec \text{now} \}, w^*, \lambda w \lambda t. \text{age}(w)(t)(\mathbf{A}) \geq 18y)$
 $= \exists t[t \supset t^* \{ \prec \text{now} \} \wedge \text{age}(w^*)(t)(\mathbf{A}) \geq 18y]$

Extrapolating from (73), an *enough* claim should establish that its subject's measure of the matrix adjective is at least d_n at reference time. Assuming that *enough* complements share temporal specification with the matrix clause derives the interpretation in (74).³⁴ The relevant set of modal alternatives is now projected from world-time pairs, on the assumption that the set of premises selected by a conversational background will differ over time, even in a single world.

- (74) $\llbracket \text{Amira était assez grande pour boire} \rrbracket^{w^*, t^*}$
 $= \llbracket \text{PST}(\text{IPF}(\sqrt{\text{Amira be old enough to drink}})) \rrbracket^{w^*, t^*}$
 $= \text{ON}(t^* \{ \prec \text{now} \}, w^*, \lambda w \lambda t. \text{age}(w)(t)(\mathbf{A}) \geq d_n)$
 $= \exists t[t \supset t^* \{ \prec \text{now} \} \wedge \text{age}(w^*)(t^*)(\mathbf{A}) \geq d_n]$, where
 $d_n = \max(\{d : \forall w \in \text{DN}(w^*, t^*)[\exists e[\tau(e) \subseteq t^* \wedge \text{drink}(w)(e)(\mathbf{A})]] \rightarrow \text{age}(w)(\tau(e)(\mathbf{A})) \geq d\})$

We saw in §3.4 (examples 57a-57c) that an *enough* complement can carry a distinct temporal specification from the matrix clause (unlike lexical implicatives; Wurmbrand 2014). The matrix-complement relationship is not unconstrained, however: as (75) illustrates, *enough* claims do not permit their complements to precede the matrix reference time.

³³Here, $\text{swim}(w^*)(e)(\mathbf{M})$ is shorthand for a representation like $\text{swim}(w^*)(e) \wedge \text{Agent}(e)(\mathbf{M})$.

³⁴A reviewer notes that, by introducing worlds in which Amira's age (and thus date of birth) varies, this example forces us to consider a set of alternatives which diverge prior to the modal projection time. This would be a problem if the circumstantial modal base for this example necessarily contained *historical* alternatives (cf. Thomason 1984) to w^* at time t^* (which must share a history with w^* until t^*), but is unproblematic if the base simply fixes certain facts of w^* that obtain at time t^* , crucially excluding Amira's actual age.

(75) #Ludo was old enough in 2024 to vote in 2020.³⁵

The evaluation time of an *enough* complement may be specified directly (via temporal adverbials) or implicitly determined from embedded material (such as definite descriptions like *the race*); where context does not provide an appropriate parameter, I assume that complement time will default to that of the matrix. The potential for matrix-complement temporal separation is captured in (76) via the introduction of a forward-expanded temporal interval in the specification of d_n ; for any interval $t \in T$, I use $t_{\rightarrow}$ to indicate the interval that begins at the earliest moment of t and extends indefinitely into the future.

$$(76) \quad \llbracket \text{enough} \rrbracket^{w,t} := \lambda P \lambda Q \lambda x. Q(w)(t)(x)(d_n), \\ d_n = \max(\{d : \forall w' \in \text{OPT}_{f,g}(w,t) [\exists e [\tau(e) \subseteq t_{\rightarrow} \wedge P(w)(e)(x) \rightarrow Q(w)(\tau(e))(x)(d)]]\})$$

This updates lexical entry (34) from §3.1 to specify that the matrix adjective is measured at a particular temporal interval and compared to a value of d_n calculated by considering appropriately-timed instances of complement realization in the set of optimal alternatives.

4.2 Aspectual coercion

We now derive the following interpretation for a perfective, dispositional *enough* construction:

$$(77) \quad \llbracket \text{Juno a été assez rapide pour gagner la course} \rrbracket^{w^*,t^*} \\ = \llbracket \text{PST}(\text{PFV}(\sqrt{\text{Juno be fast enough to win the race}})) \rrbracket^{w^*,t^*} \\ = \text{IN}(t^* \{ \prec \text{now} \}, w^*, \lambda w \lambda t. \text{speed}(w)(t)(J) \geq d_n) \\ = \exists t [t \subseteq t^* \{ \prec \text{now} \} \wedge \text{speed}(w^*)(t)(J) \geq d_n], \text{ where} \\ d_n = \max(\{d : \forall w \in \text{CR}(w^*,t^*) [\exists e [\tau(e) \subseteq t^*_{\rightarrow} \wedge \text{win}(w)(\iota x. \text{race}(x))(e)(J) \rightarrow \text{speed}(w)(t^*_{\rightarrow})(J) \geq d]]\})$$

(77) establishes that the reference time t^* contains an interval at which Juno had the capacity for the circumstantially necessary race-winning speed d_n . If we assume that the race in question is at matrix time, (77) establishes that she had this capacity during the race. This will not produce the observed implicative interpretation, however, since nothing in the interaction between the matrix predicate and the perfective (as defined in 70a) forces the stative capacity attribution to be reinterpreted as a (complement-causing) manifestation of Juno's speed. This is where **aspectual coercion** enters the picture.

Stative predicates are known to resist straightforward composition with perfective markers. This can result in uninterpretability, but it more frequently produces event (rather than state) descriptions in which the input predicate has been augmented or enriched in some way. *Inchoative* readings for perfectly-marked states are a well-known instance of this: for instance, perfective (78) situates an event of Nur becoming sad (going from not being sad to being sad) within reference time t , rather than simply locating an interval verifying Nur's sadness in t (as predicted by 70a).

- (78) Soudain, Nur a été triste.
Suddenly, Nur has be.PP sad.
'Suddenly, Nur became sad / began to feel sad.'

³⁵This example improves if a perfect auxiliary is added to the complement: i.e., *Ludo was old enough in 2024 to have voted in 2020*. Assuming that a perfect marker shifts its preadjacent backwards with respect to its own evaluation time, this is compatible with the claim that complement time (ostensibly, the result state of a prior preadjacent event) must be no earlier than the matrix; with a perfect complement, the claim would mean something like *Ludo was old enough (at $t^* \subseteq 2024$) to (at or after t^*) be in the result state of having voted in 2020*. See Nadathur (2023a, pp.143–147) for more detailed discussion of the temporal facts.

As discussed in §3.4, aspectual coercion results from the semantic incompatibility of a given input predicate with an aspectual operator. In the current setup, the problem in (78) can be formalized as one of type-level mismatch if we assume that perfectives select for eventive inputs: ³⁶

$$(79) \quad \llbracket \text{PFV} \rrbracket^{w,t} := \lambda P_{vt}. \text{IN}(t, w, P) \quad (\text{eventive-selecting perfective})$$

Successful composition can occur as long as the mismatch is repairable. Thus, the interpretation of (78) plausibly results from mapping the input stative to a predicate of closely related *inchoative* (or transition-into) events for Nur’s sadness. Formally, composition requires a bridging mechanism—a *coercion operator*—to be inserted between the predicate and aspect, as shown below: (80) offers a possible implementation for inchoative coercion (but see also Bary 2009).

$$(80) \quad \llbracket \text{INCHO} \rrbracket = \lambda w \lambda P_{it} \lambda e. \exists t_1, t_2 [t_1 \prec t_2 \wedge t_1, t_2 \subseteq \tau(e) \wedge \neg P(w)(t_1) \wedge P(w)(t_2)]$$

$$(81) \quad \begin{aligned} \llbracket \text{Nur a été triste} \rrbracket^{w^*, t^*} &= \llbracket \text{PST}(\text{PFV}(\text{INCHO}(\sqrt{\text{Nur be sad}})) \rrbracket^{w^*, t^*} \\ &= \text{IN}(t^* \{ \prec \text{now} \}, w^*, \text{INCHO}(\text{sad}(\text{N}))) \\ &= \exists e [\tau(e) \subseteq t^* \{ \prec \text{now} \} \wedge \exists t_1, t_2 [t_1 \prec t_2 \wedge t_1, t_2 \subseteq \tau(e) \wedge \neg \text{sad}(w^*)(t_1)(\text{N}) \wedge \text{sad}(w^*)(t_2)(\text{N})]] \end{aligned}$$

Where coercion is triggered, the success of reinterpretation is constrained both by selection from a small set of available coercion operators, as well as by the contextual coherence of the result (de Swart, 1998). For instance, while the punctual adverbial *soudain* (‘suddenly’) in (78) privileges an inchoative reading, the same underlying stative can elsewhere give rise to a *complexive* (bounded) interpretation: (82) conveys that Nur’s sadness both began and ended during the reference period. Complexive interpretations can be derived via MAX, as shown in (82). ³⁷

$$(82) \quad \begin{aligned} &\text{Hier, Nur a été triste.} \quad \sim \text{PST}(\text{PFV}(\text{MAX}(\sqrt{\text{Nur be sad}}))) \\ &\text{Yesterday, Nur has be.PP sad.} \\ &\text{‘Yesterday, Nur was-PFV sad.’} \\ &\text{Observed/actual: Yesterday contained a maximal period of Nur being sad.} \\ &\text{a. } \llbracket \text{MAX} \rrbracket := \lambda w \lambda P_{it} \lambda e. P(w)(\tau(e)) \wedge \forall t' [\tau(e) \subset t' \rightarrow \neg P(w)(t')] \\ &\text{b. } \llbracket (82) \rrbracket^{w^*, \text{yst}} = \exists e [\tau(e) \subseteq \text{yst} \{ \prec \text{now} \} \wedge \text{sad}(w^*)(\tau(e))(\text{N}) \wedge \forall t [\tau(e) \subset t' \rightarrow \neg \text{sad}(w^*)(t)(\text{N})]] \end{aligned}$$

Coercion can fail in contexts where no available operator produces a plausible result. For instance, perfective marking on *être grand* (‘be tall’) is odd (although reportedly not entirely unacceptable) in the presence of a punctual adverbial, since height is typically regarded as a stable (or gradually changing) property. A complexive reading is more easily accommodated: with the modifier *dans son enfance* (‘in his childhood’), (83) suggests that the state of Nima qualifying as tall (relative to a standard of tallness for people in his age group) did not persist into adulthood.

$$(83) \quad \begin{aligned} &(?)\text{Soudain} / \text{Dans son enfance, Nima a été grand.} \\ &(?)\text{Suddenly} / \text{In his childhood, Nima has be.PP tall.} \\ &\text{Inchoative: ‘(?)Suddenly Nima became tall’} \\ &\text{Complexive: ‘As a child, Nima was-PFV tall.’} \end{aligned}$$

³⁶Much of the literature treats stative predicates as a subclass of eventuality predicates. On this view, formal mismatch can then be achieved by defining PFV to select for *vt* predicates with specific lexical aspectual properties.

³⁷MAX follows a proposal from Bary. An alternative approach might treat MAX as the inverse to INCHO, so that it selects a transition-out event for a stative predicate (rather than a maximal instantiation of the predicate); this would still license the state-termination inference that are characteristic of complexive interpretations.

Inchoative and complexive aspectual coercion can explain the observed (C) interpretations for (43a) (see also 44, p.19) and (45a-i), respectively. As we have seen, these interpretations do not affect implicativity since they need not affect the likelihood of a pragmatically-driven sufficiency inference of the type discussion in §3.2.³⁸

- (43a) Amira a été assez grande pour boire de l'alcool, mais elle n'en a
 'Amira has be.PP enough old for drink.INF of the-alcohol, but she NEG-of.it has
 pas bu.
 NEG drink.PP
 'Amira was-PFV (became) old enough to drink alcohol, but she did not drink-PFV alcohol.'
- (45a-i) Nima a été assez grand pour s'enfuir, mais il ne s'est pas enfui.
 Nima has be.PP enough tall for REFL-escape.PP, but he NEG REFL-is NEG escape.PP
 'Nima was-PFV tall enough to escape, but he did not escape-PFV.'

4.3 Evidential coercion for dynamic properties

If disposition attributions are at base stative, they will only compose with a selective perfective aspect by means of coercion. Neither INCHO nor MAX will derive the manifested interpretations discussed in §3.5, suggesting that the effects we are after involve a distinct coercion operator.

- (61a) Juno a été rapide. ~ PST(PFV(INST([√]Juno be fast)))
 Juno has be.PP fast.
 'Juno was-PFV (did something) fast (#but she didn't do-PFV anything).'

I refer to the target process as **evidential** coercion, borrowing a term from Fernald (1999). Descriptively, evidential coercion maps dispositional statives to predicates of *evidentiary* actions—i.e., actions that can provide evidence supporting the stative attribution by allowing direct observation of the property in question. If Juno's having speed *d* (for instance) is understood to describe her capacity for actions characterized by speeds of at least *d*, then a performance of this speed—e.g., an event of Juno running at speed at least *d*—is precisely the type of event which might be used to justify the stative attribution, and is thus a plausible output of evidential coercion.

This form of coercion has received less attention than INCHO and MAX, but it is by no means unattested. Goldsmith and Woisetschlaeger (1982, p.85) observe the target effects for dispositions in the English progressive, noting that “where the non-progressive [(84a)] characterizes a person, the progressive [(84b)] reports on behaviour”. The ‘behavioural’ reading in (84b) is ostensibly triggered by an eventive-selecting progressive operator (see Moens 1987), and describes Alvin engaging in volitional action which would justify (84a) if it turns out to align with his standard practice.

- (84) a. Alvin is polite. (as a general fact about Alvin)
 b. Alvin is being polite (e.g., by giving up his subway seat)

Bary (2009) also makes note of progressive-triggered coercion, arguing that progressives compose only with statives if there is some pragmatically-available action which warrants description in terms

³⁸To the extent that the R pattern finds (43a) contradictory, the coercion approach must posit the existence of a coercion operator distinct from INCHO or MAX which is preferred to INCHO in (43a) and which will introduce a relationship of practical sufficiency between Amira's age and her consumption of alcohol. Homer's (2021)'s *actualistic* coercion, discussed in §4.5 seems to be the right sort of operator to capture the R effects.

of the original property. Although she does not make this explicit, the property in question must then be one which can characterize events (or at least pairs of individuals and events). De Swart (1998, p.383) cites a similar operator in progressive contexts, but does not offer a formal definition.

Fernald (1999) observes that behavioural readings for stative predicates like *be polite* and *be generous* (so-called *dispositional evaluatives*; see also Demonte 2019) also occur in non-progressive contexts, shown in (85). The temporal adverbs of quantification in these sentences introduce episodicity, intuitively quantifying over events in which the subject is “behaving [...] in a manner consistent with having the property denoted by the [predicate]”. Fernald contrasts statives that are acceptable in these temporal contexts with predicates that are odd—such as those in (86)—in terms of the degree to which the underlying predicate is associated with “standard behaviours” over which the subject “potentially has control” (pp.52–55).³⁹

- | | |
|--|---|
| <p>(85) a. Zaphod is rarely clever.
b. Pickwick is often pedantic.</p> | <p>(86) a. ??Nima is sometimes tall.
b. ??Torvald is often Norwegian.</p> |
|--|---|

The contrast in (85)-(86) maps onto the one identified in §3.4. I suggest that the same form of coercion is at work in both cases: evidential coercion is available as a repair mechanism in episodic (eventive-selecting) contexts for just those statives which are conceptually dependent on actions or events. Homer (2021), who is the first (see Homer 2011) to my knowledge to discuss a manifesting form of coercion in the presence of overt perfectivity (as opposed to the progressive and event-counting contexts in 84 and 85) arrives at a similar conclusion, characterizing reinterpretation candidates as predicates which license a modal entailment or presupposition. His proposal, for an **actualistic** form of aspectual coercion, works to derive the desired manifestation readings for dynamic properties, but turns out to also predict implicative interpretations for *enough* constructions which are non-implicative on the C pattern of judgements; I set this aside here but return to it in §4.5, where I suggest that a form of coercion closer to Homer’s may in fact be responsible for the R pattern of judgements.

The conceptual dependence between dispositional and action cuts two ways: evidential coercion requires an inherent association between a stative property and some pattern of behaviour, and this association is in turn a consequence of the stative characterization’s ontological reliance on the existence of evidentiary events. In other words, while a lexical representation like (72) offers a suitable analysis for a static predicate, a parallel representation would be both inadequate and misleading in the dispositional case.

$$(72) \llbracket \text{old} \rrbracket := \lambda w \lambda t \lambda d \lambda x. \text{age}(w)(t)(x) \geq d$$

A more appropriate treatment of *be fast* (for instance) might be along the lines of (87). Here, the stative speed of individual x over interval t is expressed in terms of an expectation that x can perform actions of a “fast” character (see also Pustejovsky, 1995). Per (87), an individual attribution of speed is made with respect to a particular kind of behaviour (Q), reflecting the intuition that *x is fast* does not require a capacity for speed in all actions taken by x , but only in some (contextually-relevant) behavioural arena (running, mathematical calculation, etc). The first line in (87) establishes a minimum evidence requirement (one fast performance of Q by x during the

³⁹To the extent that there are behaviours that are considered stereotypically Norwegian, a behavioral reading for (86b) (on which Torvald frequently engages in typically Norwegian activities) is marginal but not impossible. See also Martin and Piñón (2020) on the analysis of denominal French verbs which describe ‘stereotypical’ behaviours associated with the denotata of the stative predicates from which they are derived.

reference interval t), and the second is intended to express x 's regularly-distributed tendency for speed in doing Q ; $\mathcal{R}_t^Q$ introduces a contextually-determined regular partition of t (cf. Deo 2009), whose cell size/partition measure depends on both Q and t .⁴⁰

$$(87) \quad \llbracket \text{fast} \rrbracket^{w,t} = \lambda x \lambda d. \exists Q_{evt} [\text{IN}(t, w, \lambda w \lambda e. Q(w)(e)(x) \wedge \text{speed}(e) \geq d)) \\ \wedge \forall t' \in \mathcal{R}_t^Q [\exists w' \in \text{CR}(w, t) [\exists e. \tau(e) \circ t' \wedge Q(w')(e)(x) \wedge \text{speed}(e) \geq d]]]$$

The speed specification in (87) constitutes a lexically-specified behavioural characterization peculiar to *fast*; other dynamic stative are associated with different *characters*. Letting $\text{Char}(S)$ stand in for this specification, I propose (88) as a rough template for the stative, individual-level use of dynamic capacity predicates.⁴¹

$$(88) \quad \llbracket S_{\text{disp}} \rrbracket^{w,t} = \lambda x. \exists Q_{vt} [\text{IN}(t, w, \lambda w \lambda e. Q(w)(e)(x) \wedge \text{Char}(S)(e)) \\ \wedge \forall t' \in \mathcal{R}_t^Q [\exists w' \in \text{CR}(w, t) [\exists e. \tau(e) \circ t' \wedge Q(w')(e)(x) \wedge \text{Char}(S)(e)]]]$$

Definition (89a) sets out the conditions under which an eventive predicate P can qualify as evidence for a disposition S with respect to an individual argument x : it must be the case that all events of type P have the character of the input stative S , involve the subject of S in a suitable participant (**Part**; **Agent** or **Theme**) role, and that all such events coincide with an interval over which S holds of x . I define the evidential coercion operator INST in terms of Evid : INST maps $S(x)$ to a unique, contextually-salient evidentiary predicate. The results of evidential coercion are shown for the perfective of *être drôle* ('be funny') in (90).

$$(89) \quad \text{a. } \text{Evid}(P_{vt}, S_{\text{disp}}(x)) \text{ iff } \forall e, w [P(w)(e) \rightarrow [\text{Char}(S)(e) \wedge \text{Part}(e)(x) \wedge S(w)(\tau(e))(x)] \\ \wedge \forall w, t [\forall t' \in \mathcal{R}_t^P [\exists e. \tau(e) \circ t' \wedge P(w)(e) \rightarrow S(w)(t)(x)]]]$$

$$\text{b. } \llbracket \text{INST} \rrbracket := \lambda w \lambda S_{it} \lambda e. (\iota P_{vt}. \text{Evid}(P, S(x)))(w)(e)$$

$$(90) \quad \text{a. } \text{Khalid a été drôle.} \quad \text{PST(PFV(INST}(\sqrt{\text{Khalid be funny}})) \\ \text{Khalid has be.PP funny.}$$

'Khalid was-PFV (did something that was) funny.'

$$\text{b. } \llbracket (90a) \rrbracket^{w^*, t^*} = \text{IN}(t^* \{ \prec \text{now} \}, w^*, \text{INST}(\llbracket \text{funny} \rrbracket (\text{K}))) \\ = \exists e [\tau(e) \subseteq t^* \{ \prec \text{now} \} \wedge (\iota P. \text{Evid}(P, \llbracket \text{funny} \rrbracket (\text{K}))) (w^*)(e)]$$

Based on these truth conditions, (90a) will be true just in case Khalid has (within reference time) performed some action—telling a good joke, doing an impression—which evidences his funniness as at or above some contextual threshold; i.e., if the instantiated evaluation world event e is funny (has the mirth-provoking event **Character** associated with *funniness*) and if events of this sort, regularly instantiated over an interval t , would license a stative predication of Khalid's funniness at t .

⁴⁰The tendency for speed is defined weakly here, based on my subjective impression that one does not need to regularly perform a behaviour Q at speed in order to be considered fast at doing Q : it is enough to be *capable* of regular fast performances of Q . A reviewer inquires whether the speed characterization of the event in (87) should instead be considered as a measurement of some movement of the participant object over the course of the event: this seems reasonable to me and compatible with (87) insofar as the event over which speed can be measured is lexicalized as an event involving an individual object.

⁴¹The first, minimum-evidence conjunct should most likely be omitted for certain dynamic properties, such as non-agentive dispositions like *cost* (see §4.5), which do not require any witnesses in order to hold over a certain period. The definition of the *evidence* relation in (89a) will not be affected by this.

4.4 Implicativity through evidential coercion

Implicative interpretations. Returning to *enough* constructions, recall that circumstantial claims with dispositional adjectives exhibit ‘fully implicative’ behaviour (on the C pattern) only where additional conditions are met: the matrix adjective must plausibly characterize the proximate cause of the complement eventuality, and the temporal specification of the complement must coincide with that of the matrix clause. These features support a contextual presupposition of causal sufficiency between a manifestation of the matrix disposition and the realization of the *enough* complement. As a result, applying evidential coercion will license an implicative entailment.

Example (91) illustrates. Under evidential coercion, (91) returns an event in which Juno evidences speed of d_n or higher. Given the causal presupposition associated with (91) (updated to the temporal framework in 92), together with the assumption that the complement’s time (i.e., the time of the referenced race) overlaps with matrix time, (91) describes the execution of a causally-sufficient condition for Juno to win the race, and thus derives the desired implicative inference.

$$\begin{aligned}
 (91) \quad & \llbracket \text{Juno a été rapide pour gagner la course} \rrbracket^{w^*, t^*} \\
 &= \llbracket \text{PST}(\text{PFV}(\text{INST}(\sqrt{\text{Juno be fast enough to win the race}})) \rrbracket^{w^*, t^*} \\
 &= \text{IN}(t^* \{ \prec \text{now} \}, w^*, \text{INST}(\llbracket \text{fast} \rrbracket (J)(d_n))) \\
 &= \exists e. [\tau(e) \subseteq t^* \{ \prec \text{now} \} \wedge (\iota P. \text{Evid}(P, \llbracket \text{fast} \rrbracket (J)(d_n)))(w^*)(e)], \text{ where} \\
 & d_n = \max(\{d : \forall w \in \text{CR}(w^*, t^*) [\exists e [\tau(e) \subseteq t^* \wedge \text{win}(w)(\iota x. \text{race}(x))(e)(J) \rightarrow \llbracket \text{fast} \rrbracket^{w, \tau(e)} (J)(d)]]\}) \\
 & \text{There is a reference time event of Juno performing a salient action (i.e., running) at or} \\
 & \text{above the necessary race-winning speed.}
 \end{aligned}$$

$$\begin{aligned}
 (92) \quad & \text{Causal necessity and sufficiency for dynamic enough constructions:} \\
 & \forall w \in \text{CS}(w^*, t^*) [\exists e [\tau(e) \subseteq t^* \wedge \text{INST}(\llbracket \text{fast} \rrbracket (J)(d_n))(w)(e) \leftrightarrow \text{win}(w)(\iota x. \text{race}(x))(e)(J)]]], \text{ where } d_n \\
 & \text{is as defined above.}
 \end{aligned}$$

Coercion also takes place in the negative case in (93). Here, the stative predicate attributes speed lower than d_n to Juno. Evidential coercion will still return a salient evidentiary action, but in this case one which evidences speed that is insufficient for Juno to win the race. Since the causally normal developments of a below-threshold speed manifestation do not include winning, we derive the desired entailment to complement negation.

$$\begin{aligned}
 (93) \quad & \llbracket \text{Juno n’a pas été rapide pour gagner la course} \rrbracket^{w^*, t^*} \\
 &= \llbracket \text{PST}(\text{PFV}(\text{INST}(\sqrt{\text{NOT}(\text{Juno be fast enough to win the race}})) \rrbracket^{w^*, t^*} \\
 &= \text{IN}(t^* \{ \prec \text{now} \}, w^*, \text{INST}(\neg \llbracket \text{fast} \rrbracket (J)(d_n))) \\
 &= \exists e. [\tau(e) \subseteq t^* \{ \prec \text{now} \} \wedge (\iota P : \text{Evid}(P, \neg \llbracket \text{fast} \rrbracket (J)(d_n)))(w^*)(e)], \text{ where } d_n \text{ is as in (91)} \\
 & \text{There is a reference time event of Juno performing a salient action (i.e., running) at less} \\
 & \text{than the necessary race-winning speed.}
 \end{aligned}$$

English does not overtly mark aspect. Nevertheless, as previously discussed, past-tense uses of dispositions are systematically ambiguous between eventive and stative readings, suggesting that coercion processes can occur in the absence of overt triggers (see also Fernald 1999). Taking into account the correlation between episodic contexts and evidential interpretations (see 59 in §3.5), the emergence of complement inferences for dispositional English *enough* claims can be attributed to the presence of a *covert* perfective operator: we observe particularly strong implicative inferences when temporal context privileges an episodic (perfective) interpretation for the matrix attribution of a dispositional and circumstantial English *enough* claim.

- (94) a. At the track meet this morning, Juno was fast enough to win her race.
 $\leadsto$ *Juno won the race.*
 b. At the track meet this morning, Juno was not fast enough to win her race.
 $\leadsto$ *Juno did not win the race.*

Crucially, however, since nothing in the observable structure of an English *enough* claim forces a speaker to indicate their choice between eventivity and stativity, a uncoerced (non-evidential) reading is always available for claims like (94a)-(94b), though it may be pragmatically dispreferred. The result is thus as suggested in §3.5: the defeasibility contrast between French and English *enough* claims is not due to an underlying difference in the semantics of *enough* and *assez*, but results instead from a contrast in the visibility of aspectual markers in the two languages.

Lastly, although evidential coercion appears to be the default repair mechanism for dispositional predicates, it is not the only option for composition with the perfective. Selection of a coercion operator can be influenced by context: thus, if it is indeed aspectual coercion which precipitates the implicative inferences in (91)-(93), contexts which privilege non-evidential coercion should license non-implicative readings for perfective, dispositional *enough* claims. The prediction is upheld: (95a) and (95b) are both felicitous and non-implicative.⁴²

- (95) a. Olga a soudain été assez forte pour soulever un frigo, mais elle n'a
 Olga has suddenly be.PP enough strong for lift.INF a fridge, but she NEG-has
 pas essayé de le faire.
 NEG try.PP of it do.INF.
 'Olga suddenly became strong enough to lift a fridge, but she did not try to do it.'
 b. Pendant deux heures ce matin, Olga a été assez forte pour soulever un
 during two hours DEM morning, Olga has be.PP enough strong for lift.INF a
 frigo, mais elle ne l'a jamais fait.
 fridge, but she NEG it-has never do.PP.
 'For two hours this morning, Olga had enough strength to lift a fridge, but she never did it.'

Modification by *soudain* ('suddenly') in (95a) privileges inchoativity, while the bounding *pendant deux heures ce matin* ('for two hours this morning') modifier in (95b) privileges a complexive interpretation. These data support an evidential-coercion based approach to *enough*'s aspect sensitive complement entailments since they show that implicativity evaporates with other forms of coercion.

Non-implicative interpretations. Absent perfectivity, non-implicative interpretations for positive dispositional *enough* constructions need little further explication. A stative disposition composes directly with the imperfective marker (covert in English) to produce a reference time state in which the subject is asserted to be capable of manifesting the matrix property to the threshold degree d_n . Like (91), (96) is associated with the background causal assumption in (92), but causal sufficiency remains 'inactive' since (96) does not describe a race time manifestation of Juno's speed.

- (96) $\llbracket \text{PST}(\text{IPF}(\sqrt{\text{Juno be fast enough to win the race}})) \rrbracket^{w^*, t^*}$
 $= \exists t[t \supset t^* \{ \prec \text{now} \} \wedge \exists Q_{vt}[\text{IN}(t, w, \lambda w \lambda e. Q(w)(e)(x) \wedge \text{speed}(e) \geq d_n)]$

⁴²These examples are based on similar data involving ability modals from Homer (2011) and Mari and Martin (2009), respectively.

$$\wedge \forall t' \in \mathcal{R}_t^Q [\exists w \in \text{CR}(w^*, t^*) [\exists e. \tau(e) \circ t' \wedge Q(w)(e)(x) \wedge \text{speed}(e) \geq d_n]]$$

The reference interval verifies a state of Juno having the capacity to perform salient actions at speeds at or above d_n .

The more complex cases are those which involve matrix negation, since compatibility with a non-implicative interpretation appears to go against the expectation set by the baseline necessity condition of *enough* (here, that the subject’s capacity for the threshold degree of the matrix adjective is circumstantially necessary for the complement realization). As discussed in §3.5, complement-compatible uses of negated circumstantial *enough* can arise with both progressive and non-progressive readings: the progressive effect can be explained in terms of evidential coercion, while the treatment of dispositions offered in §4.3 accounts directly for the other case.

We have already seen that progressive marking can trigger evidential readings for dispositions (see 84b). This suggests that the progressive patterns with the perfective in selecting for eventives, although it shares the ‘including’ semantics of the imperfective as set out in (70b). Progressive interpretations are well-attested for imperfectively-marked French claims, supporting the hypothesis that the French *imparfait* can be realized as PRG (97a) in certain contexts, such as where aspect-supplementing temporal information supports an episodic perspective. I assume that the *imparfait* is elsewhere realized as a stative-selecting IPF (so that PRG and IPF divide the labour attributed to IPF in 70b). Inserting PRG between tense and the (negated) *enough* radical in (98) produces a past in-progress manifestation of below-threshold speed.

$$(97) \quad \text{a. } \llbracket \text{PRG} \rrbracket^{w,t} := \lambda P_{vt}. \text{ON}(t, w, P) \quad \text{b. } \llbracket \text{IPF} \rrbracket^{w,t} := \lambda P_{vt}. \text{ON}(t, w, P)$$

$$(98) \quad \llbracket \text{Juno n'était pas assez rapide pour gagner la course} \rrbracket^{w^*, t^*}$$

$$= \llbracket \text{PST}(\text{PRG}(\text{INST}(\text{NOT}(\vee \text{Juno be fast enough to win the race})))) \rrbracket^{w^*, t^*}$$

$$= \text{ON}(t^* \{ \prec \text{now} \}, w^*, \text{INST}(\neg \llbracket \text{fast} \rrbracket (J)(d_n)))$$

$$= \exists e. [\tau(e) \supset t^* \{ \prec \text{now} \} \wedge (\iota P : \text{Evid}(P, \neg \llbracket \text{fast} \rrbracket (J)(d_n)))(w^*)(e)], \text{ where } d_n \text{ is as in (91)}$$

At reference time, Juno was in the middle of performing a salient action (i.e., running) at less than the necessary race-winning speed.

Since the causal background for this *enough* claim establishes the causal necessity of a threshold manifestation for complement realization (see 92), Juno’s performance in (98) leads to the expectation that she will not win the race. Crucially, however, this prediction is based on the reference time perspective, which is—by definition—fully contained within Juno’s performance: the relevant facts can subsequently change. In order for this to happen however, some ongoing process must change or otherwise be interrupted, and this is precisely what happens in a non-implicative example like (63a): although Juno’s win is unexpected at reference time, a later change in speed permits her to win after all. Evidential but non-implicative readings are also possible for progressively-interpreted positive *enough* claims, despite an evidential reading: in (99), Juno’s ongoing performance of race-winning speed can be disrupted after the reference interval, but prior to the end of the race, allowing an outcome in which she fails to triumph.

$$(63a) \quad (\text{Au début,}) \text{ Juno n'était pas assez rapide pour gagner la course, mais}$$

$$(\text{at-the start,}) \text{ Juno NEG-be.PST.IPF NEG enough fast for win.INF the race, but}$$

$$\text{dans le dernier tour, elle a couru beaucoup plus vite et a finalement}$$

$$\text{in the final lap, she has run.PP much more quickly and has finally}$$

$$\text{réussi à gagner.}$$

$$\text{succeed.PP at win.INF.}$$

‘(At the start of the race,) Juno was-PRG not (running) fast enough to win, but in the final lap she ran-PFV much faster and ultimately managed-PFV to win.’

- (99) (Au début,) Juno était assez rapide pour gagner la course, mais dans le (at-the start,) Juno be.PST.IPF enough fast for win.INF the race, but in the dernier tour, elle a couru plus lentement et a finalement terminé à la troisième final lap, she has run.PP more slowly and has finally end.PP at the third place.
position.

‘(At the start of the race,) Juno was-PRG (running) fast enough to win, but in the final lap she ran-PFV slower and eventually finished in third place.’

Negated dispositional *enough* claims are also compatible with complement realization in non-progressive contexts, as seen in (65b).

- (65b) Juno n’était pas assez rapide pour gagner la course, mais (à cause d’un Juno NEG-be.PST.IPF NEG enough fast for win.INF the race, but (from cause of-a médicament) elle a couru inhabituellement vite à cette fois-là et elle a drug) she has run.PP unusually quickly on DEM time-DIST and she has réussi à gagner.
succeed.PP at win.INF.

‘Juno was-IPF not (ordinarily) fast enough to win the race, but (because of a drug) she ran unusually fast on that occasion and managed to win.’

This is rather less puzzling than it was before, as the analysis proposed in §4.3 allows an individual x to verify disposition S over an interval t just in case x exhibits a regularly-distributed tendency for acts with the character specified by S . Interpreted statively, then, the *enough* claim in (65a) requires Juno to have a reference-time disposition for running slower than the necessary threshold speed d_n , but this is compatible with the possibility that she occasionally runs at or above speed d_n during t . If one such exceptional performance happens to coincide with the relevant race event, the truth of (65b) does not rule out an exceptional win.

Comparison with lexical implicatives. As discussed in §3.6, lexical implicatives and *enough* predicates differ with respect to the content and aspectual class of their assertions: this difference derives a contrast in the aspect sensitivity of their complement inferences. Lexical implicatives’ complement entailments persist under non-progressive imperfectivity because implicative verbs are at base eventive, asserting the realization of a necessary/sufficient action for their complements. If IPF selects for stative inputs (as in 97b), imperfectly-marked lexical implicatives must undergo eventive-to-stative coercion. Based on the intuitive interpretation of non-progressive but imperfective *réussir* (‘manage’, ‘succeed’), the default reinterpretation appears to be **habitual**:

- (100) Nur réussissait à résoudre les énigmes.
Nur succeed.PST.IPF at solve.INF the riddles.
‘Nur (typically) managed to solve riddles.’

(101a) provides a definition for HAB, on which it maps an eventive input P to a predicate of times during which P is instantiated at a majority of relevant intervals (i.e., those which satisfy some salient predication R) in a contextually-determined regular partition ($\mathcal{R}_t^c$) of the input time t ; I

assume that R minimally picks up any presuppositions associated with P (cf. Schubert and Pelletier 1989 on GEN). The treatment of *manage* (*réussir*) in (101b) is based on (15) (§2.2.2, p.8), but updated for temporal interpretation: following observations from Karttunen (1971) and Wurmbrand (2014), *manage* (like other lexical implicatives) restricts its complement to coincide with the matrix evaluation time. (101c) spells out the results of inserting HAB between the implicative sentence radical and the imperfective marker.

- (101) a. $\llbracket \text{HAB} \rrbracket := \lambda w \lambda t \lambda R_{it} \lambda P_{vt} . \text{MOST } t' [t' \in \mathcal{R}_t^c \wedge R(w)(t')] [\text{IN}(t', w, P)]$
 b. $\llbracket \text{manage} \rrbracket^{w,t} := \lambda e \lambda P_{vt} \lambda x . (\iota A_{vt} . \forall w' \in \text{CS}(w, t) [\text{IN}(t, w', A(x)) \leftrightarrow \text{IN}(t, w', P(x))]) (w)(e)$
 c. $\llbracket (100) \rrbracket^{w^*, t^*} = \llbracket \text{PST}(\text{IPF}(\text{HAB}(\sqrt{\text{Nur manage to solve riddles}})) \rrbracket^{w^*, t^*}$
 $= \exists t [t \supset t^* \{ \prec \text{now} \}$
 $\wedge \text{MOST } t' [t' \in \mathcal{R}_t^c \wedge \iota A . \forall w \in \text{CS}(w^*, t') [\text{IN}(t', w, A(x)) \leftrightarrow \text{IN}(t', w', \text{solve-riddle}(\text{N}))]]$
 $\quad \quad \quad [\text{IN}(t', w, A(x))]$

On the habitual reading in (101c), *manage* (*réussir*) requires that the past reference time is contained in a larger interval during which most occasions where Nur had a causally necessary and sufficient causing action for riddle-solving were occasions in which she took that action. Each of these causing actions is necessarily linked to a realization of the *manage* complement, making not just the prerequisite but also its result constitutive of Nur's 'habit'. Non-universal quantification in HAB renders (101c) compatible with a small number of complement failures, but directly contradicts any claim that establishes Nur's regular or uniform failure to solve riddles, as shown in (102).

- (102) a. Nur (generally) managed to solve riddles, { but she couldn't solve this one / #but she mostly/always failed to solve them }
 b. Nur réussissait à résoudre les énigmes, { mais cette énigme-là était Morgan succeed.PST.IPF at solve.INF the riddles, { but DEM riddle-DIST be.PST.IPF trop difficile / #mais (d'habitude) elle n'en résolvait jamais. }
 too difficult / #but (of-habit) she NEG-them solve.PST.IPF never.
 'Morgan managed-IPF to solve riddles, {but that riddle was too difficult / #but she (usually) never solved-IPF them.'

No contradiction arises where *manage* is replaced with a dispositional *enough* predicate, as in (103a)-(103b), because an imperfective interpretation here requires only that Nur has the capacity to behave with degree d_n of cleverness, but does not entail an instantiation of the proximate, cleverness-characterized cause of riddle-solving in any particular situation.

- (103) a. Nur was clever enough to solve riddles, but she never did it.
 b. Morgan était assez intelligente pour résoudre les énigmes, mais elle Morgan be.PST.IPF enough clever for solve.INF the riddles, but she n'en résolvait jamais.
 NEG-them solve.PST.IPF never.
 'Morgan was-IPF clever enough to solve riddles, but she never solved-IPF them.'

Where progressive readings are accepted for lexical implicatives, they pattern with progressively-interpreted dispositional *enough* claims in failing to license complement entailments:⁴³

⁴³While I find progressives of English implicatives somewhat marked, albeit not impossible (as do several consul-

- (104) a. Mika was managing to open the door when a noise scared her and she dropped the key.
 b. γ Noémie Leclerc réussissait enfin / était enfin en train de
 Noémie Leclerc manage-PST.IPF finally / be.PST.IPF finally in process of
 réussir à monter le projet de Xavier Beauvois quand Élise Formain est
 manage.INF to mount the project of Xavier Beauvois when Élise Formain is
 venu tout foutre en l’air.
 come.PP all screw in the-air.
 ‘Noémie Leclerc was finally managing to get Xavier Beauvois’s project off the ground
 when Élise Formain came along and screwed everything up.’

This is again as predicted: an eventive implicative can compose with the progressive to license readings on which the causal prerequisite is in progress at reference time. However, as per the imperfective paradox, and as the examples in (104) show, the entailed causing event can be disrupted before completion, permitting non-implicativity despite the lexical semantics of the underlying verb.

4.5 An alternative: Homer’s (2021) *actualistic* coercion

By capitalizing on a conceptual contrast between static and dynamic properties, evidential coercion works both to derive empirically-observed manifested readings for monoclausal dynamic property attributions, as well as to explain the C pattern of judgements for *enough* implicativity. In this section, I discuss an alternative, due to Homer (2011, 2021), which derives predictions that seem to align with the R pattern of *enough/assez* judgements.

Homer points out that predicates like *être intelligent/e* (‘be intelligent’) license entailments to certain events or actions under perfective (but not imperfective) marking. His proposal—for an **actualistic** form of coercion—thus aims at deriving the same type of reading as evidential coercion.

- (105) a. Jean a été intelligent.
 Jean has be.PP intelligent
 ‘Jean was-PFV intelligent.’
 $\rightarrow$ *Jean acted with intelligence.*
 b. Jean n’a pas été intelligent
 Jean NEG-has NEG be.PP intelligent
 ‘Jean was-PFV not intelligent’
 $\rightarrow$ *Jean acted without intelligence*

Homer assumes that the perfective selects *quantized* (telic) rather than eventive predicates, characterizing candidates for actualistic coercion as predicates licensing a modal entailment or presupposition.⁴⁴ In (106) (Homer, 2021, p.38), *R* represents the input predicate, while *Q* is contextually

tants), they appear to be fine for many speakers. I thank an anonymous reviewer for the French example (104b) which demonstrates the compatibility (and non-implicativity) of *réussir* with both a progressively-interpreted imperfective and the periphrastic progressive *être en train de* (‘be in the process of’).

⁴⁴The quantization restriction is common in the coercion literature, often motivated by the observation that perfective processes have bounded (complexive) readings. The default reading of (1) suggests the presence of MAX: this is expected if PFV selects for quantized inputs, but not if it is compatible with all eventives.

- (1) Juno a couru ce matin.
 Juno has run.PP this morning.
 ‘Juno ran-PFV this morning.’ (*This morning contained a maximal/completed running eventuality by Juno.*)

(1) requires an alternative explanation on an account (as in §4.2) on which PFV composes with processes. I cannot properly explore the motivations for such an account here, but the simplest approach might fold complexive meaning (as in 82a) into the contribution of PFV (cf. Krifka 1998). Coupled with a treatment of statives as temporal predicates,

supplied; the second conjunct of ensures that Q is quantized. The final conjunct constrains the relationship of Q and R , using ∂ (Beaver and Krahmer, 2001) to require that ACT is defined only where the *strongest* modal entailment of R is necessary and sufficient for a reference time Q -event.

$$(106) \quad \llbracket \text{ACT} \rrbracket := \lambda w \lambda t \lambda R_{vt} \lambda Q_{vt} \lambda e. Q(e)(t)(w) \wedge \forall e_1 [e_1 \sqsubset e \rightarrow \neg Q(e_1)] \wedge [\exists e_2 : R(e_2)(t)(w)] \\ \wedge \partial(\exists e_3 [Q(e_3)(t)(w) \wedge \tau(e_3) \subseteq t] \leftrightarrow \Omega),$$

with Ω the strongest entailment of $\exists e : R(e)(t)(w)$ of the form $[\text{Mod}(\exists e_4 : P(e_4)(t))]$,
 $\text{Mod} \in \{\diamond, \square\}$ and P some predicate of eventualities.

(107) shows how the interpretations in (105) are derived using ACT; I suppress the quantization conjunct and treat θ_i as the contextual threshold for positive intelligence. Following Homer, I treat Ω as the presupposed possibility that Jean is the agent of some event.⁴⁵ As long as this possibility is necessary and sufficient for an event of type Q at reference time (see 107c), ACT will instantiate a reference-time Q -event in both (107a) and (107b). In (107a), Q is accompanied by a reference-time state of Jean’s intelligence meeting or exceeding θ_i ; (107b) entails a reference-time state of Jean’s intelligence failing to reach θ_i (when combined with Ω and 107c). Entailments to intelligent and unintelligent action (respectively) are derived as long as the best candidate for Q is some deliberate action by Jean (i.e., Ω ’s prejacent).⁴⁶ This interpretation should arise only where any other conditions for Jean’s action (a suitable task to be undertaken or a decision to be made, Jean’s being awake at the time, etc) are taken for granted. As these conditions are necessarily satisfied in any context where he chooses to act, presuppositional ACT is *ipso facto* compatible with contexts where the derived interpretations are observed.

(107) Modal presupposition of *Jean être intelligent/e*: *Jean can act agentively/deliberately*

$$\Omega = \exists w \in \text{CR}(w^*) [\exists e [\text{Agent}(\text{J})(e)(w) \wedge \tau(e) \circ t^*]]$$

- a. $\llbracket (105a) \rrbracket^{w^*, t^*} = \llbracket \text{PST}(\text{PFV}(\text{ACT}(Q_{\text{qn}})(\sqrt{\text{Jean be intelligent}}))) \rrbracket^{w^*, t^*}$
 $= \exists e_1 [\tau(e_1) \subseteq t^* \{ \prec \text{now} \} \wedge Q(e_1)(w^*)] \wedge \exists e_2 [\text{smart}(\text{J})(e_2)(w^*) \geq \theta_i \wedge \tau(e_2) \circ t^*]$
There is a Q-event and a state of Jean being at least θ_i intelligent at reference time.
- b. $\llbracket (105b) \rrbracket^{w^*, t^*} = \llbracket \text{PFV}(\text{NEG}(\text{PFV}(\text{ACT}(Q_{\text{qn}})(\sqrt{\text{Jean be intelligent}}))) \rrbracket^{w^*, t^*}$
 $= \neg \exists e_1 [\tau(e_1) \subseteq t^* \{ \prec \text{now} \} \wedge Q(e_1)(w^*)] \vee \neg \exists e_2 [\text{smart}(\text{J})(e_2)(w^*) \geq \theta_i \wedge \tau(e_2) \circ t]$
There is no Q-event or Jean is less than θ_i intelligent at reference time.
- c. *Presupposition supplied by ACT*: $\exists e_3 [Q(e_3)(w^*) \wedge \tau(e_3) \subseteq t^*] \leftrightarrow \Omega$
The reference-time possibility of Jean’s agentive action is necessary and sufficient for a complete Q-event

this move preserves a selectional restriction based on semantic type (rather than on finer-grained aspectual properties), and is compatible with a partitive theory of aspect (Filip, 2000; Filip and Rothstein, 2005; Altshuler, 2014).

⁴⁵Since Homer assumes that negation scopes above grammatical aspect (and thus operates on the output of ACT; see 107b), Ω must be presupposed if ACT is to produce the desired entailments under both matrix polarities. (Homer, 2021, pp.30,40) makes this explicit for the predicate *coûter* (‘cost’), and I model (107) on his analysis of this predicate.

⁴⁶Formally, the intelligence states and the derived actions in (107) are instantiated with slightly different relations to reference time. This is a consequence of Homer’s temporal architecture, on which (vP) predicates of eventualities are automatically instantiated as overlapping reference time; grammatical aspects (e.g., PFV) override this only for free event variables in their scope. As a result, (107) is compatible with non-overlapping instantiations of Jean’s action and his stative intelligence, but this seems to be an unintended technical issue, and I set it aside. A broader question is whether or not the relative freedom of Q in ACT leads to overprediction for actualized interpretations; I will not attempt to adjudicate this here, but see, e.g., Nadathur (2023a, pp.217–219, 244–247) for further discussion.

It should be clear from (107) that actualistic coercion will derive implicative inferences from dynamic (and circumstantial) *enough* constructions under perfective marking: the only difference between a claim like (107a) and—for instance—the perfective claim that Jean is intelligent enough to solve a riddle is that in the latter case Jean’s intelligence should be compared to the threshold degree d_n required for riddle-solving instead of to a contextual standard for intelligence. Where Jean’s exercise of cleverness is additionally taken to be causally necessary and causally sufficient for (reference-time) riddle solving (see §3.4, see also 92), the interpretation derived in (107a) will license a (causal) entailment to the *enough* complement; under negation, the entailment that Jean acted with insufficient intelligence will derive a causal entailment to complement failure.

This is not the only way in which *enough* implicativity may be derived using ACT: there is an alternative, which does not require reference to the background condition of causal sufficiency/necessity discussed in §3.4. As a consequence of the necessity semantics for *enough*, positive *enough* claims entail the possibility of their complements. If this entailment plays the role of Ω , actualistic coercion must presuppose that the possibility of the *enough* complement is necessary and sufficient for the coerced event Q at reference time. If Q is identified (as above, and in several of Homer’s examples) with the prejacent of Ω , a positive perfective *enough* claim will derive both that its complement occurred and that the subject had the necessary degree of the matrix adjective; a negative claim will establish that its complement did not occur and that the subject lacked the necessary degree of the matrix adjective.⁴⁷ This is shown for (91) and its negation in (108).

(108) Modal entailment of *Juno être assez rapide pour gagner*: *Juno can win*

$$\Omega = \exists w \in \text{CR}(w^*)[\exists e[\text{win}(\text{J})(e)(w) \wedge \tau(e) \circ t^*]]$$

- a. $\llbracket \text{PST}(\text{PFV}(\text{ACT}(Q_{\text{qn}})(\sqrt{\text{Juno be fast enough to win}})) \rrbracket^{w^*, t^*}$
 $= \exists e_1[\tau(e_1) \subseteq t^* \{ \prec \text{now} \} \wedge Q(e_1)(w^*)] \wedge \exists e_2[\text{speed}(\text{J})(e_2)(w^*) \geq d_n \wedge \tau(e_2) \circ t^*],$
 Where $d_n = \max(\{d : \forall w \in \text{CR}(w^*)[\exists e_3[\text{win}(\text{J})(e_3)(w) \rightarrow \text{speed}(\text{J})(e_3)(w) \geq d]]\})$
There is a Q-event and a state of Juno having at least speed d_n at reference time
- b. $\llbracket \text{PST}(\text{NEG}(\text{PFV}(\text{ACT}(Q_{\text{qn}})(\sqrt{\text{Juno be fast enough to win}}))) \rrbracket^{w^*, t^*}$
 $= \neg \exists e_1[\tau(e_1) \subseteq t^* \{ \prec \text{now} \} \wedge Q(e_1)(w^*)] \vee \neg \exists e_2[\text{speed}(\text{J})(e_2)(w^*) \geq d_n \wedge \tau(e_2) \circ t^*]$
There is no Q-event or Juno has less than speed d_n at reference time
 $\rightarrow$ *There is no Q-event and Juno has less than speed d_n at reference time*
 (when 108c is taken into account)
- c. *Presupposition introduced by ACT*: $\exists e_4[Q(e_4)(w^*) \wedge \tau(e_4) \subseteq t^*] \leftrightarrow \Omega$

In principle, adjudicating between (108) and a derivation closer to (107) comes down to determining whether the modal entailment of *enough* is stronger than a presupposition of potential action (which must be lexically associated with all dispositional predicates in order to derive manifested readings for their monoclausal perfectives). I am not certain how to make this determination, since the candidates for Ω do not diverge in modal force (and Homer does not discuss such a scenario). Both routes derive the target inferences, but make slightly different predictions with respect to what is required of the sentential subject. Crucially, since Ω in (108) is entailed rather than presupposed, (108b) differs from (107b) in precluding a Q -event. But this means that (108b) should be compatible

⁴⁷More precisely, a negative claim requires that at least one of these conditions holds, but since ACT’s presupposition (108c) equates Q with the possibility of the complement (Ω), and since Ω holds if the subject has at least degree d_n of the matrix adjective, the truth conditions in (108b) can only be satisfied when both disjuncts are verified.

with a lack of action on the part of the sentential subject—i.e., with the claim that Juno did not even try to win the race; (107b), on the other hand, entails (indeed, presupposes) some action (the event Q). My consultants find a negated and perfectly-marked dispositional *enough* claim odd if its subject did not even try to achieve the complement—the imperfective is preferred in such a context—which perhaps suggests that the derivational route in (107) (which relies on both the lexical semantics of the matrix predicate and a contextually-supplied causal background rather than the modal semantics of *enough*) is to be preferred here.

A derivation like (108), however, remains relevant for *enough* constructions with static predicates (which by themselves do not license modal entailments). The modal semantics of *enough* makes both (109b) and (109c) candidates for actualistic coercion. The predicted interpretations are paraphrased in (109); once again, these truth conditions will derive implicative entailments just in case (a) the best candidate for Q is the prejacent of the modal entailment Ω (i.e., the *enough* complement) and (b) the reference time possibility of escape (Ω) is regarded as necessary and sufficient for its realization. In other words, the condition of practical sufficiency which is—per §3.2–§3.4—needed to derive positive implicative inferences via pragmatic reasoning is here supplied presuppositionally by ACT, offering a link between grammatical aspect and implicativity.

(109) Modal entailment of *Nima être assez grand pour s’enfuir*: *Nima can escape*

- a. *Presupposition supplied by ACT*: The possibility of Nima’s escape is necessary and sufficient at reference time for a Q -event (for Q a contextually-relevant quantized predicate).
- b. Nima a été assez grand pour s’enfuir.
Nima has be.PP enough tall for REFL-escape.INF
‘Nima was-PFV tall enough to escape.’
Actualistic truth conditions: There is a Q -event and Nima is at least d_n tall at reference time, where d_n is the necessary height for escaping.
- c. Nima n’a pas été assez grand pour s’enfuir.
Nima NEG-has NEG be.PP enough tall for REFL-escape.INF
‘Nima was-PFV not tall enough to escape.’
Actualistic truth conditions: There is no Q -event and Nima is less than d_n tall at reference time.

Since ACT as defined in (106) does not restrict the modal entailment Ω to be of any particular flavour, actualistic coercion is also predicted to apply to deontic *enough* constructions: this is schematized for *Amira a été assez grande pour boire* (‘Amira was-PFV old enough to drink’) and its negation in (110). As in (109) the assumption of practical sufficiency needed to link complement legality to Amira’s (attempt at) complement realization (see §3.2) is supplied by the presupposition of ACT, producing implicative inferences by grammatically backgrounding the pragmatic context required for an implicative interpretation of the underlying *enough* claim.⁴⁸

(110) Modal entailment of *Amira être assez vieille pour boire*: *Amira is permitted to drink*

⁴⁸Where the matrix predicate of a deontic *enough* claim is dispositional—as in the chess club example in (3b) or the Boston registration example (62b)—we are in the same situation as with dispositional and circumstantial *enough*. Identifying Ω with the action-presupposition of the matrix predicate will derive predicate-characterization action (under both positive and negative matrix polarities; see 107), but requires a background inference of practical sufficiency (linking this action causally to attempted complement realization) in order to derive implicativity. Identifying Ω instead with the deontic possibility of the *enough* complement will derive implicativity directly in perfective contexts, but does not commit the speaker to predicate-characterized action in the negative case.

- a. *Presupposition supplied by ACT*: The legality of Amira’s drinking is necessary and sufficient at reference time for a Q_{qn} event
- b. $PST(PFV(ACT(Q)(\vee \text{Amira be old enough to drink})))$
There is a Q -event and Amira is at least age d_n at reference time, where d_n is the legal minimum drinking age.
- c. $PST(NEG(PFV(ACT(Q)(\vee \text{Amira be old enough to drink}))))$
There is no Q -event and Amira’s age is less than d_n at reference time.

Since ACT in some sense builds in the context required for pragmatically-derived implicativity, the difference between an account based on evidential coercion and a (more permissive) actualistic approach does not lie in whether implicative readings are possible at all for examples like (109) and (110), but rather in the means—purely pragmatic or coercion-driven—by which these inferences are derived. Deciding between the two approaches is thus rather tricky, but seems to come down to two things: (a) whether the implicative inference arises by default in cases like (109) and (110) (more plausible on the actualistic route as a result of presupposition accommodation) or (b) whether the implicative interpretation co-occurs with inferences that suggest the application of an alternative (inchoative or complexive) form of coercion. The latter is necessary on the evidential route since pragmatically-derived implicativity does not on its own resolve the problem of type mismatch between the matrix property attribution of an *enough* claim and a selective perfective operator.

As far as I can tell, these two points align with the ways in which the R and C judgements for *enough* diverge. The reported R judgement reflects a default for implicativity in an example with the form of (110b) (see (43a) in §3.2), to the extent that an alternative inchoative repair is strongly dispreferred, but my consultants’ judgements seem to reflect inchoativity (i.e., a reference-time change in age) even when complement realization is (pragmatically) inferred. Similarly, my consultants associate perfective claims like (109b)-(109c) with inferred changes in the relationship between Nima’s height and the height needed for escape, suggesting that implicativity for them is derived independently of the process of coercion that licenses aspectual composition; if R speakers perceive no such inference on the implicative reading, this supports the view that these judgements correspond to two different forms of perfective-triggered coercion.⁴⁹

Another possibility, which would bring the two sets of judgements closer together, is that C speakers do employ actualistic coercion but apply it only to dispositions (i.e., predicates constructed along the lines proposed in 87), strongly dispreferring it for the reinterpretation of predicates which are not ontologically dependent on a modal entailment. This would involve a variant of ACT that is more restricted than (106) about the nature of Ω , while R speakers employ the more permissive (106). Dispositional-ACT speakers would then still rely on a causal background to link the entailed content of a perfective, actualized, dispositional *enough* to complement entailments, so the calculation and distribution of implicative inferences would be much the same on this approach as described in §4.4. I leave a detailed investigation of these possibilities—and a more precise characterization of the more permissive R judgements—for future work.

To the extent that aspectual coercion is an interpretation-facilitating repair operation and

⁴⁹A similar point is made by the R judgement which finds the negated imperfective example in (45a-ii) (*Nima was-IPF tall enough to escape*, see p.19) has to have a nonimplicative interpretation. This is not possible on the C pattern, but if we assume R speakers employ actualistic coercion (and that this form of coercion can be triggered in non-perfective but episodic contexts), the judgement is as predicted: since imperfective *enough* claims with static adjectives are, like their dispositional counterparts, candidates for actualistic coercion under a progressively-interpreted imperfective marker, we do not expect differential behaviour from these two predicate classes in this context.

thus—while grammatical—has a pragmatic basis (see also Bary 2009, pp.85–86), it is not that surprising to find individual variation in either the definition of a coercion operation or the distribution of coerced interpretations. The two different patterns of judgement discussed here reflect a split in whether truly dispositional predicates are treated differently from predicates which license contextual modal entailments, and in a sense represent two distinct ways of reasoning pragmatically: the restrictive C pattern involves reasoning about how a particular disposition attribution may be locally (episodically) justified (see also Piñón 2009) while the other is more closely associated with the pragmatics of exhaustivity (or *antiperfectivity*), and merely requires a local event in which the relevant property can be construed as instrumental, rather than one which is characterized by or directly evidences the property itself.

4.6 Revisiting the three puzzles

At the beginning of §3, I outlined three puzzles in the implicativity of *enough* claims: inference optionality, variation in inferential profile, and (aspect-sensitive) inference strength variability. §3-§4 provided empirical and formal accounts of these puzzles, which I summarize briefly below.

Optionality. Deontic (non-circumstantial) *enough* constructions lack implicative defaults, while circumstantial claims show some form of implicativity (§3.2, see also Karttunen 1971; Meier 2003). I argued that this *optionality* is predicted and explained by the necessity semantics for *enough* introduced in §3.1 (cf. von Stechow et al. 2004), once we recognize that constraints on the matrix-complement relationship in an *enough* construction are enforced only over the appropriate modal quantificational domain. Deontic quantificational domains need not include the evaluation world, so reality in these cases need not conform to *enough*'s necessity condition. However, since circumstantial modal domains do include the evaluation world, these constructions are formally predicted to exhibit (at least) a one-way, necessity-driven pattern of complement inferences.

Distribution/inferential profile. Within the circumstantial subgroup of *enough* constructions, some speakers show a contrast between ‘two-way’ and ‘one-way’ claims: the former license aspect-sensitive implicativity under both matrix polarities, and the latter only in the (necessity-driven) negative case. I suggested in §3.4 that this contrast follows from a difference between *dispositional* vs. *static* matrix adjectives, and the way in which the properties these predicates invoke are conceptually involved in the realization of the *enough* complement. Dispositional adjectives license two-way implicative inferences just in case they can be understood as invoking a property which characterizes a sufficient (proximate) cause of the *enough* complement; in combination with the necessity semantics from §3.1, this structure very nearly replicates the implicative semantic profile proposed by Nadathur (2023b), on which lexical implicatives assert the realization of a prerequisite (action) which is presupposed to be *causally necessary and sufficient* for their complements.

Inference strength. I linked the *variable strength* puzzle to the interaction of aspectual class and grammatical aspect (see §3.5); §4 formalized this proposal. I argued that the eventive interpretation of perfective dispositional *enough* constructions is a case of *aspectual coercion*, here treated as an operation which repairs type mismatches between an eventive-selecting perfective and a stative *enough* input. I suggested that the target interpretation for dispositional *enough* reflects the effect of an **evidential** form of coercion, which applies exclusively to dispositional statives in virtue of their

ontological dependence on action potentials (§4.2-§4.3). Based on this characterization of its target inputs, I offered a formal proposal for evidential coercion (INST; 89b), on which it maps a disposition to a salient predicate of evidentiary events for the underlying stative. §4.4 shows (a) how evidential coercion derives ‘full’ (two-way, entailed) implicativity when it occurs in tandem with contextual support for a two-way causal presupposition and matrix-complement temporal overlap, (b) how the actionable nature of dispositional claims predicts the non-implicativity of imperfectly-marked negated dynamic *enough*, and (c) how aspectual class explains the differential aspect sensitivity of complement inferences in *enough* vs. lexical implicative claims. Finally, in §4.5, I showed how an alternative pattern of *enough* implicativity judgements might be derived using Homer’s (2021) *actualistic* coercion, and discussed the potential connection between the two derivational routes.

5 Conclusions and outlook

This paper offers an account of the variable implicativity of *enough* constructions which draws on lexical semantics, compositional structure, and contextually-supplied causal knowledge to make good on the hypothesis that *enough*’s complement inferences, where they arise, are derived from the same semantic components as the characteristic complement entailments of lexically implicative verbs (Nadathur 2023b). Although this unified analysis vindicates to some degree Hacquard’s (2005) claim that *enough* is semantically implicative, I have shown that her original observations (see §1, §3.3) are the product of a specific and distributionally-restricted conspiracy: *enough* constructions mimic the inferential profile of lexical implicatives only where fine-grained aspectual class properties (associated with the property invoked by the matrix adjective) and contextual knowledge (about modal flavour and the causal background against which the *enough* complement is realizable) come together to recreate the complex lexical semantic structure of ‘true’ implicative verbs.

This paper makes a broader case for the relevance of causal information across the lexical, compositional, and discursive levels. The underlying idea is that causal structure of the sort that supports implicative inferences can not only be lexically referenced (as by lexical implicatives) and/or license pragmatic inferences (e.g., about narrative coherence; Hobbs 1990, a.o.), but can also bridge between the lexical and pragmatic levels, supplementing lexically-encoded information (such as the necessity presupposition of *enough*) with structured world knowledge in a manner that has observable consequences for compositional interpretation and non-defeasible entailment.

The account presented here supports a view of implicativity as a semantic phenomenon grounded in causal structure, and in particular grounded in causal structure pertaining to an individual’s ability to realize goals in view of their propensity for realizing a more immediate, intermediary causing condition. This suggests—in keeping with an intuition expressed by Bhatt (1999)—a deep connection between implicative semantics and the longstanding puzzle of **actuality inferences**, wherein past tense ability attributions imply the realization of their complements (see also Thalberg 1972). Like the implicative inferences of (dynamic, causal) *enough* constructions, ability predicates’ actuality inferences are sensitive to aspectual marking: as shown in (111), perfectly-marked French abilitative *pouvoir* (‘can’) licenses **actuality entailments**, but its imperfective alternative remains compatible with complement non-realization.⁵⁰

⁵⁰ Actuality entailments are observed across a wide range of aspect-marking languages; the literature is rather extensive. See, a.o., Bhatt (1999); Hacquard (2006); Mari and Martin (2009); Piñón (2009); Matthewson (2012); Mari (2016); Alxatib (2020); Homer (2021); Nadathur (2019, 2021); Hacquard (2020) provides a recent overview.

- (111) a. Olga **a pu** soulever cette table, #mais elle ne l’a pas fait.
 Olga has can.PP lift.INF this table, #but she NEG it-has NEG do.PP
 ‘Olga could-PFV lift this table, #but she did not do it.’
 b. Olga **pouvait** soulever cette table, mais elle ne l’a jamais fait.
 Olga can.PST.IPF lift.INF this table, but she NEG it-has never do.PP
 ‘Olga could-IPF lift this table, but she never did it.’

On the current analysis, *enough* claims are a special type of ability attribution, wherein the subject’s ability to realize the complement depends on their propensity for realizing actions characterized by the matrix property. This sets up an analogy between prerequisite-specifying lexical implicatives and *manage*, on the one hand, and *enough* and ‘pure’ ability predicates, on the other:

dare : manage :: be brave enough : be able / can_{able}

Dare and *manage* share core causal structure, but differ in prerequisite specificity; following Nadathur (2023b), this explains their shared inferential profile as well as contrasts in their appropriate use contexts. The suggestion above is that the parallels between (causal) *enough* constructions and ability attributions can likewise be explained by treating *be brave enough* as a prerequisite-specifying structural counterpart to actualizing ability predicates.

If the analysis provided here is on the right track, implicative *dare* contrasts with *be brave enough* in that the former is both eventive and prerequisite-realizing, while the latter is stative and prerequisite-permitting: this explains differences in the strength and aspect sensitivity of these predicates’ complement inferences. *Manage* is then the eventive, prerequisite-realizing alternative to a stative, prerequisite-permitting ability predicate. Extending the current account of implicativity to actuality inferences, then, requires positing that ability attributions describe a complex causal structure between some (underspecified) action and the realization of the ability target. Such an analysis makes ability claims complex causal dispositions, subject to evidential coercion when they occur in eventive-selecting perfective contexts, and thus to recreating the full semantic structure of *manage* in these environments. The idea gains some support from Bhatt (1999), who points out that the parallels between actualized ability and *manage* extend beyond complement entailment to a shared presupposition of complement non-triviality (see 112): this is explained in terms of the presupposed causal background in the implicative case (Nadathur, 2023b).

- (112) Mika { managed / did not manage / was able / was not able } to breathe normally.
 → *Breathing normally was non-trivial (difficult, unexpected, abnormal?) for Mika*

The semantic details and consequences of this informally-sketched proposal—and the corresponding success of extending a congruent analysis from lexical implicative and *enough* claims to ability attributions—are beyond the scope of this paper, and will have to be explored elsewhere. Nevertheless, while a complex, tiered structure for ability diverges from the standard linguistic (premise semantics) treatment, it is not without precedent in the philosophical literature (Brown, 1988; Belnap, 1991; Mandelkern et al., 2017, a.o.). Looking ahead, the true viability of an implicative approach to the semantics of ability will be measured not only in its success in accounting for aspect-sensitive actuality inferences, but also in its success in using complex causal structure to explain a number of long-observed peculiarities in the modal logic of ability claims as compared to other uses the possibility modal, and other flavours of possibility (Kenny, 1976, a.o.).

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